Acoustics of porous media

Lecture 3: Variational formulations and FEM discretization

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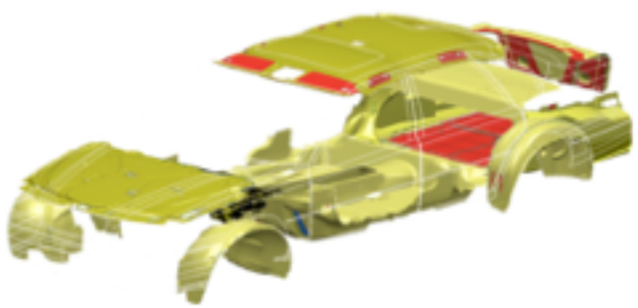
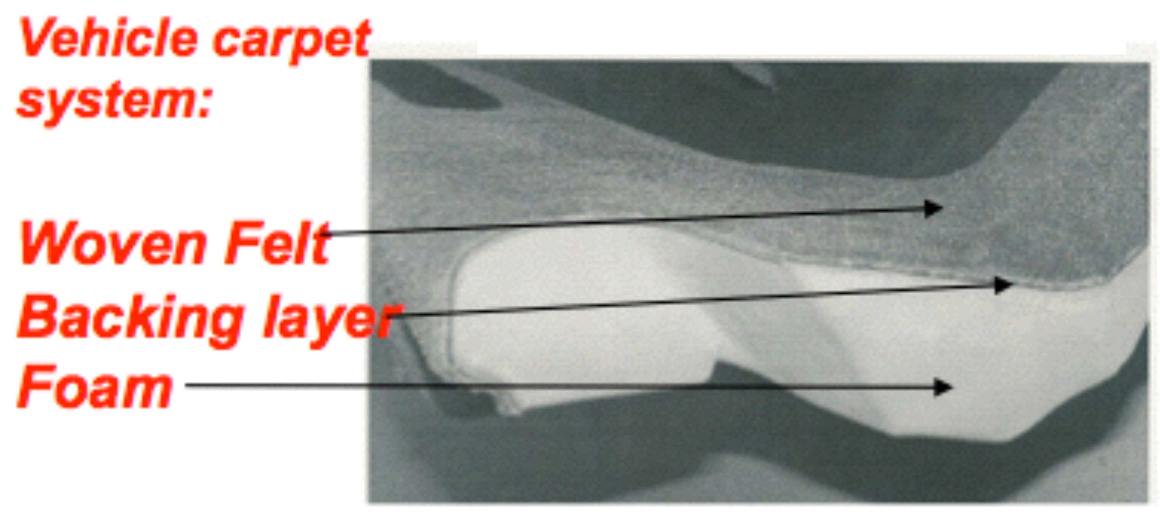
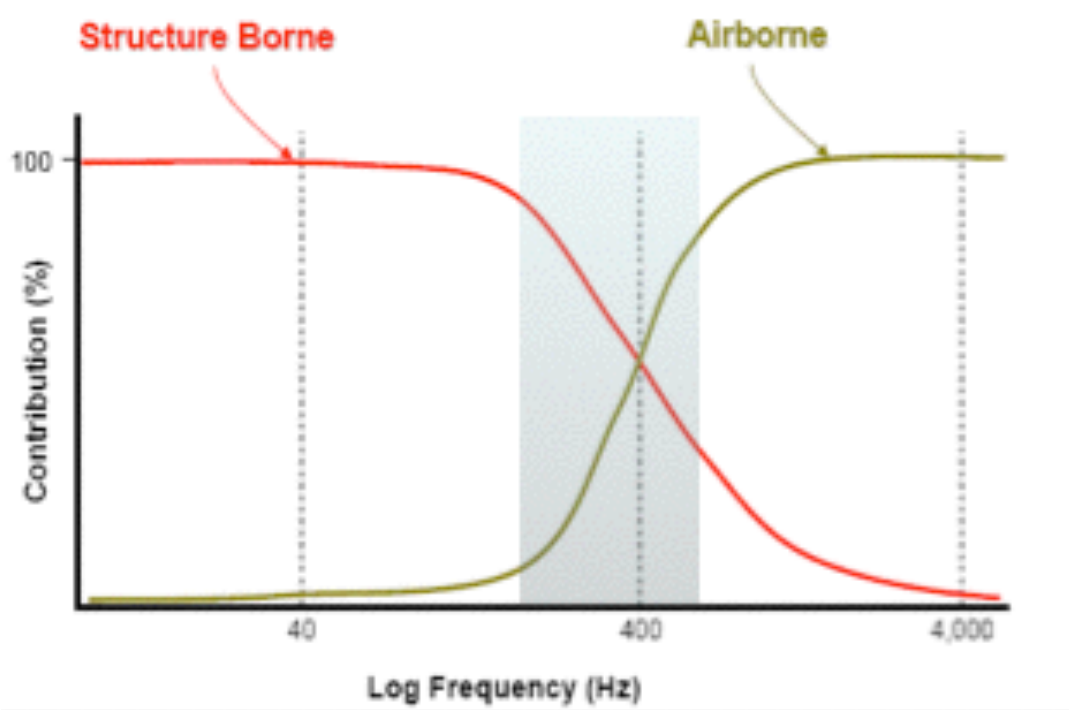
Summary of the lecture

- Variational formulations
  - Equivalent fluid - Limp
  - Biot - Displacement
  - Mixed formulation
- Coupling conditions
- Discretization
  - Mixed formulation







- Dissipation mechanisms
  - Viscoelasticity (solid - LF)
  - Viscous-thermal (fluid - HF)

1 pressure field models

Behavioral criterion quantifying the edge-constrained effects on foams in the standing wave tube

Dominic Pilon and Raymond Panneton
GAUS, Department of Mechanical Engineering, Université de Sherbrooke, Sherbrooke, Quebec J1K 2R1, Canada

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[DOI: 10.1121/1.1598193]

PACS numbers: 43.55.Ev, 43.20.Mv, 43.20.Jr [LLT]

FAE

Validity of the limp model for porous materials: A criterion based on the Biot theory

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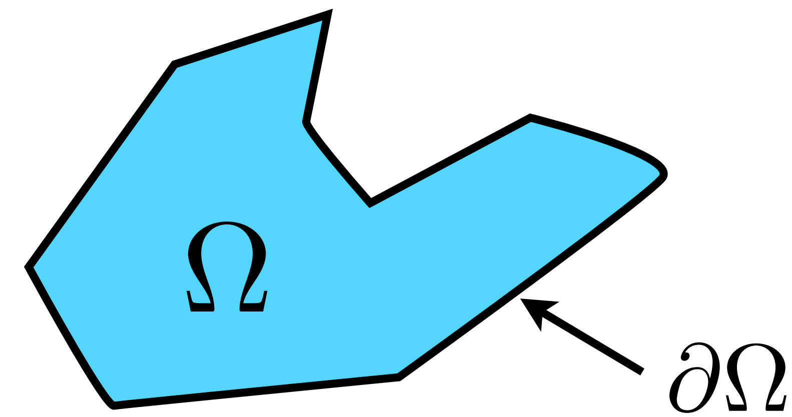

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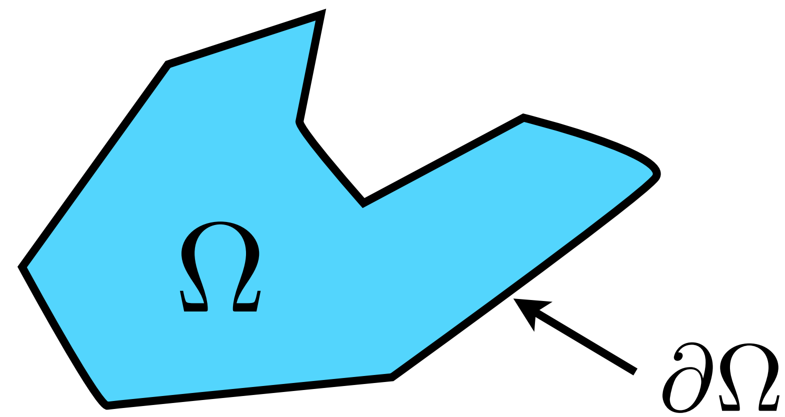

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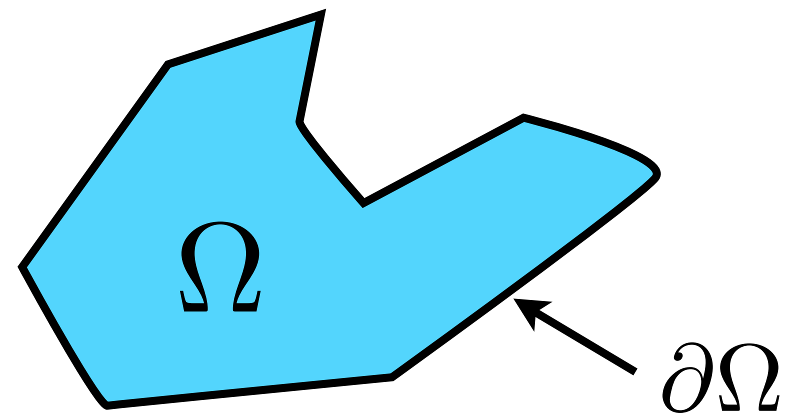

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$$= \oint_{\partial\Omega} u_n^{eq} \delta p \, \mathrm{d}\Gamma.$$

P models (Equivalent fluids,limp model)

$$\forall \delta p \in V_p(\Omega), \quad \int_\Omega \frac{\boldsymbol{\nabla} p \ \boldsymbol{\nabla} \delta\, p}{\widetilde{\rho}_{eq}\omega^2} - \frac{p \delta p}{\widetilde{K}_{eq}} \ \mathrm{d}\Omega = - \oint_{\partial\Omega} \frac{1}{\widetilde{\rho}_{eq}\omega^2} \frac{\partial p}{\partial n} \delta p \, \mathrm{d}\Gamma.$$

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$$\left(\frac{[\mathbf{H}_0]}{\widetilde{\rho}_{eq}\omega^2} - \frac{[\mathbf{Q}_0]}{\widetilde{K}_{eq}} \right) \mathbf{p} = \mathbf{u}$$

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$$\int_\Omega \nabla \Phi_{\,i} \, \nabla \Phi_{\,j} \; \mathrm{d}\Omega$$

$$\left(\frac{[\mathbf{H}_0]}{\widetilde{\rho}_{eq}\omega^2} - \frac{[\mathbf{Q}_0]}{\widetilde{K}_{eq}} \right) \mathbf{p} = \mathbf{u}$$

“Kinetic energy matrix”

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$$\int_\Omega \boldsymbol{\nabla} \Phi_i \, \boldsymbol{\nabla} \Phi_j \ \mathrm{d}\Omega \qquad \int_\Omega \Phi_i \Phi_j \ \mathrm{d}\Omega$$

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“Kinetic energy matrix”

“Compression energy matrix”

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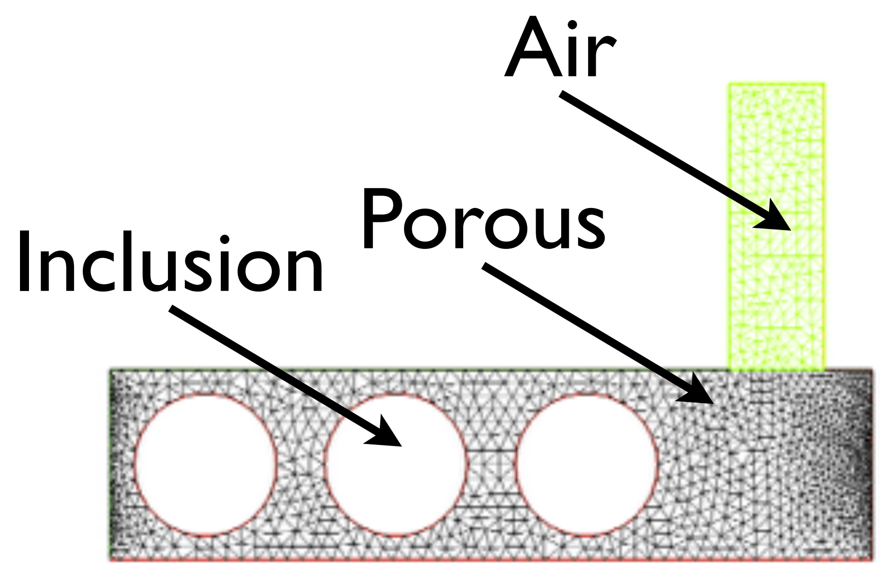
“Kinetic energy matrix”

“Compression energy matrix”

- Discretisation through linear-quadratic elements
- Complex system
- 2 real and sparse matrices
- Natural coupling with air - other equivalent fluid models

Example: metaporous materials

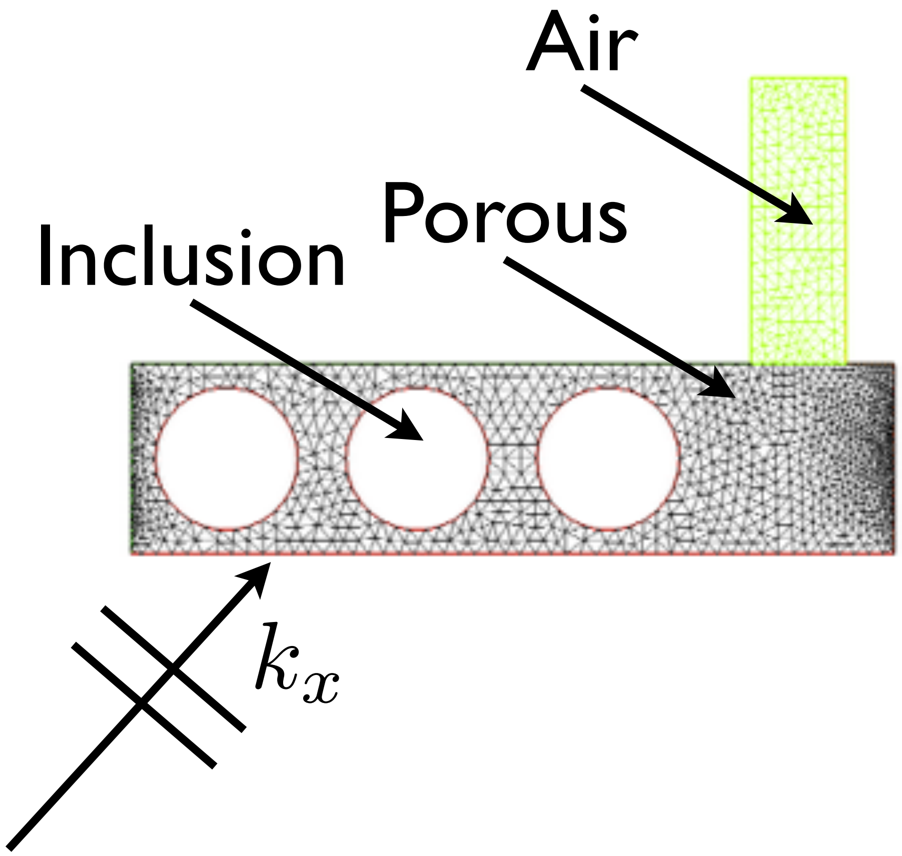


2D Bloch-periodic

Fireflex

- ✓ Thickness 2 cm
- ✓ Spatial periodicity 8 cm
- ✓ 3 inclusions (radius 7.5 mm)
- ✓ Cavity dimension: 3 cm x 1 cm

Example: metaporous materials



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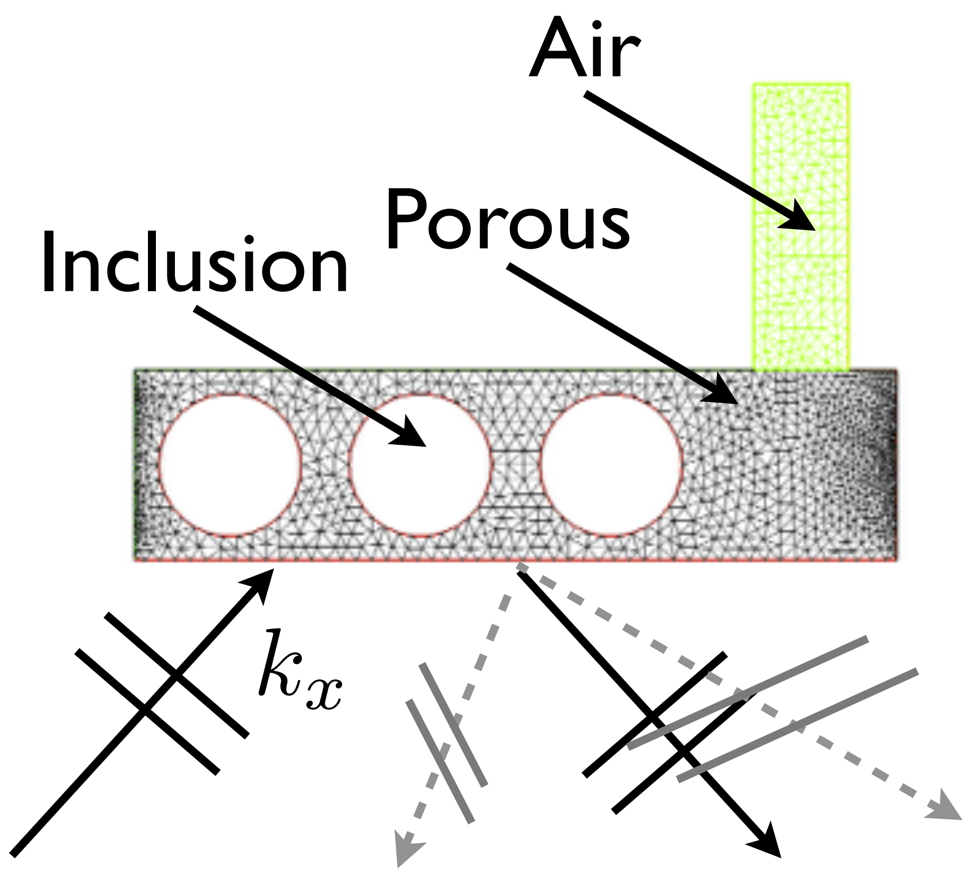

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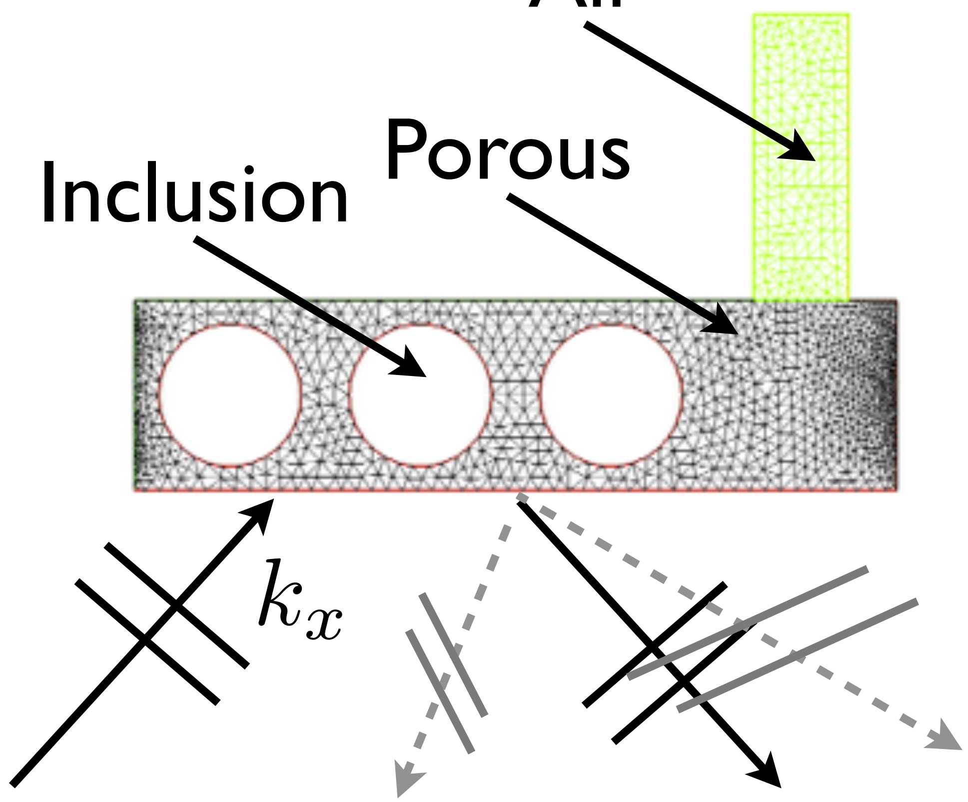

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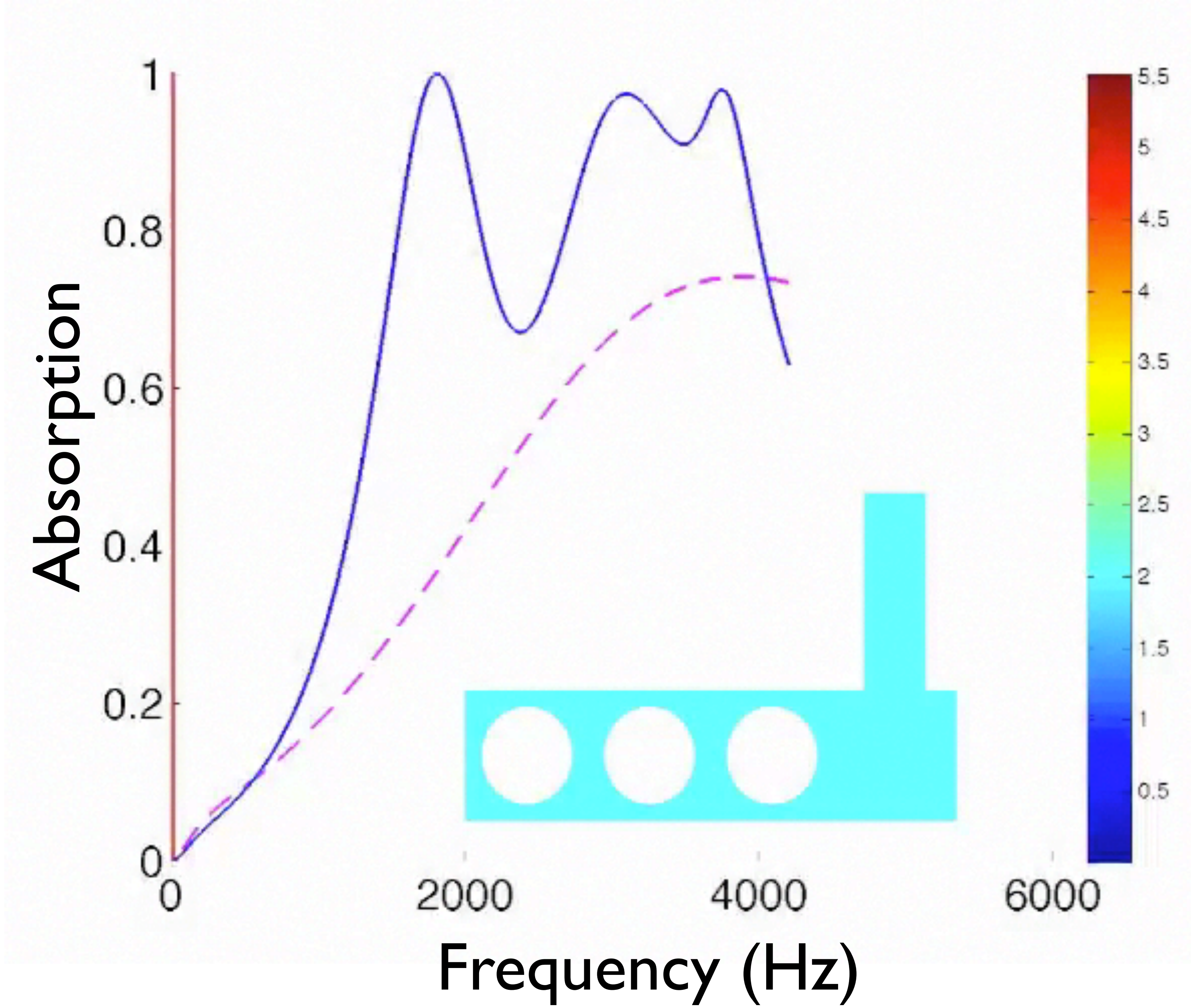

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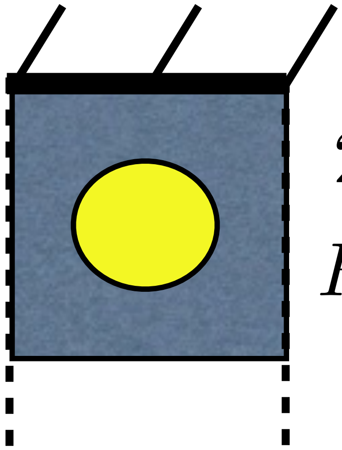
Groby *et al.* JASA **127(5)** 2010

Prediction of resonances



$2\,\mathrm{cm} \times 2\,\mathrm{cm}$

$R = 0.75\,\mathrm{cm}$

Fireflex

$$\phi = 0.96$$

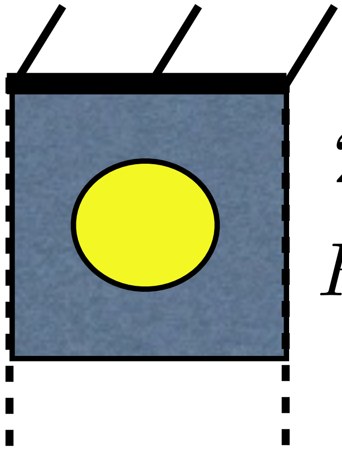
$$\alpha_\infty = 1.42$$

$$\Lambda = 180\,\mu\mathrm{m}$$

$$\Lambda' = 360\,\mu\mathrm{m}$$

$$\sigma = 8900\,\mathrm{Nsm}^{-4}$$

Prediction of resonances



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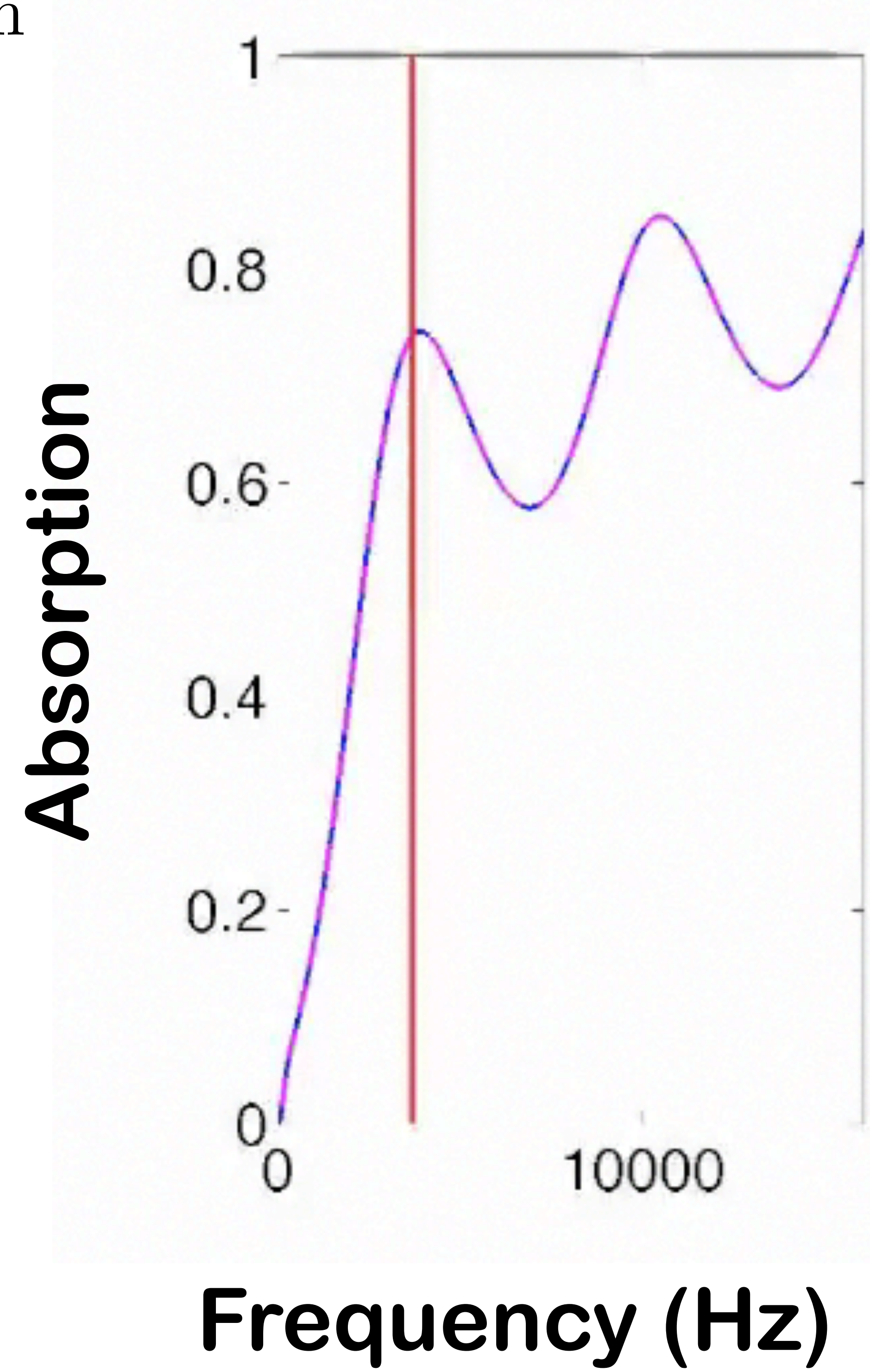
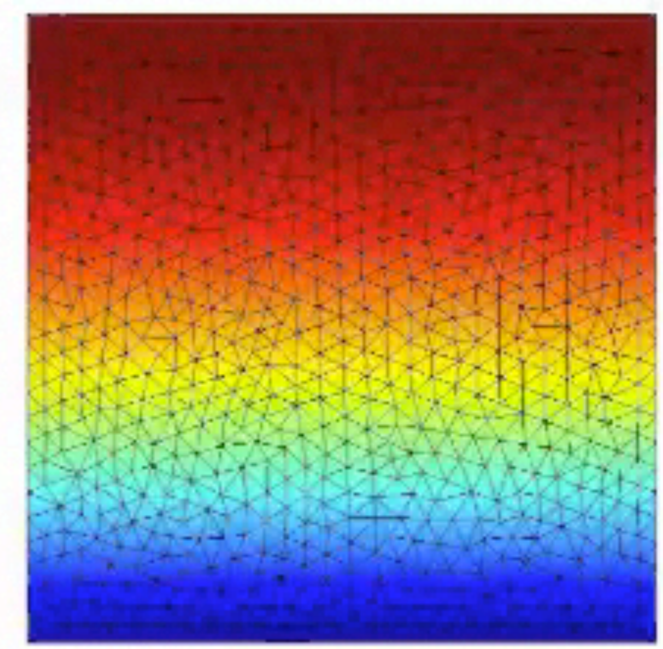
Evolution of absorption versus the size of the solid inclusion



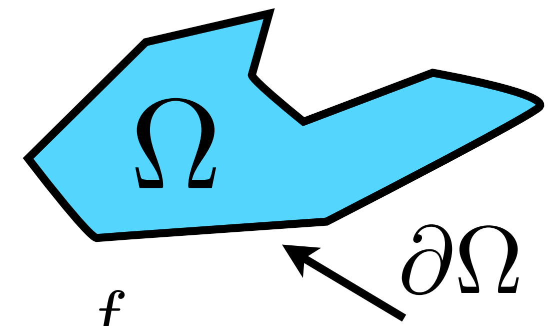


Pressure profile of the first mode

Biot original formulation

$$
\begin{aligned}
\nabla.\boldsymbol{\sigma}^s(\mathbf{u}^s,\mathbf{u}^f) &= -\omega^2\widetilde{\rho}_{11}\,\mathbf{u}^s - \omega^2\widetilde{\rho}_{12}\,\mathbf{u}^f,\\
\nabla.\boldsymbol{\sigma}^f(\mathbf{u}^s,\mathbf{u}^f) &= -\omega^2\widetilde{\rho}_{12}\,\mathbf{u}^s - \omega^2\widetilde{\rho}_{22}\,\mathbf{u}^f
\end{aligned}
$$

Biot original formulation



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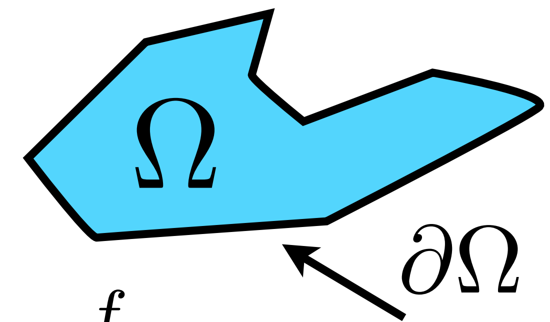
\boldsymbol{\nabla}.\boldsymbol{\sigma}^f(\mathbf{u}^s, \mathbf{u}^f) &= -\omega^2 \widetilde{\rho}_{12}\, \mathbf{u}^s - \omega^2 \widetilde{\rho}_{22}\, \mathbf{u}^f
\end{aligned}
$$

PEM: displacement formulations

Biot original formulation

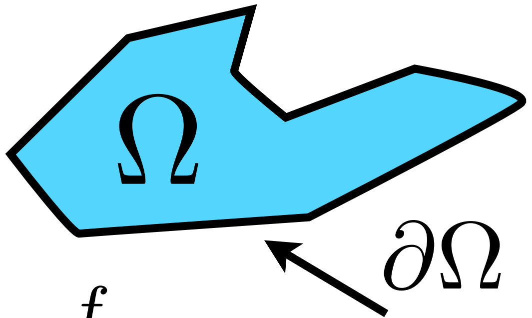


$$
\begin{aligned}
\boldsymbol{\nabla}.\boldsymbol{\sigma}^{s}(\mathbf{u}^{s},\mathbf{u}^{f}) &= -\omega^{2}\widetilde{\rho}_{11}\,\mathbf{u}^{s} - \omega^{2}\widetilde{\rho}_{12}\,\mathbf{u}^{f},\\
\boldsymbol{\nabla}.\boldsymbol{\sigma}^{f}(\mathbf{u}^{s},\mathbf{u}^{f}) &= -\omega^{2}\widetilde{\rho}_{12}\,\mathbf{u}^{s} - \omega^{2}\widetilde{\rho}_{22}\,\mathbf{u}^{f}
\end{aligned}
$$

$\mathbf{u}^{s} \in V_{s}$ (Subspace of $(H^{1}(\Omega))^{3}$ filtered by $\mathbf{u}^{s} = 0$ on $\partial\Omega_{u}^{s}$

$\mathbf{u}^{f} \in V_{f}$ (Subspace of $(H^{1}(\Omega))^{3}$ filtered by $\mathbf{u}^{f}.\mathbf{n} = 0$ on $\partial\Omega_{u}^{f}$

Biot original formulation



$$\begin{aligned}\boldsymbol{\nabla}.\boldsymbol{\sigma}^s(\mathbf{u}^s,\mathbf{u}^f) &= -\omega^2\widetilde{\rho}_{11}\,\mathbf{u}^s - \omega^2\widetilde{\rho}_{12}\,\mathbf{u}^f,\\ \boldsymbol{\nabla}.\boldsymbol{\sigma}^f(\mathbf{u}^s,\mathbf{u}^f) &= -\omega^2\widetilde{\rho}_{12}\,\mathbf{u}^s - \omega^2\widetilde{\rho}_{22}\,\mathbf{u}^f\end{aligned}$$

$\mathbf{u}^s \in V_s$ (Subspace of $(H^1(\Omega))^3$ filtered by $\mathbf{u}^s = 0$ on $\partial\Omega_u^s$

$\mathbf{u}^f \in V_f$ (Subspace of $(H^1(\Omega))^3$ filtered by $\mathbf{u}^f.\mathbf{n} = 0$ on $\partial\Omega_u^f$

Methodology (Panneton,Coyette, Bolton...)

- $(1)\times\delta\mathbf{u}^s \in V_s$
- $(2)\times\delta\mathbf{u}^f \in V_s$
- Integration on Ω, Green Riemann
- Sum of both terms

Displacement formulation: weak forms

Solid phase

$$\forall \delta \mathbf{u}^s \in V_s, \quad \int_\Omega \boldsymbol{\sigma}^s(\mathbf{u}^s, \mathbf{u}^f) : \boldsymbol{\epsilon}(\delta \mathbf{u}^s)\, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho}_{11} \mathbf{u}^s \delta \mathbf{u}^s + \widetilde{\rho}_{12} \mathbf{u}^f \delta \mathbf{u}^s \, \mathrm{d}\Omega$$

$$= \oint_{\partial\Omega} \boldsymbol{\sigma}^s(\mathbf{u}^s, \mathbf{u}^f).\mathbf{n}\delta \mathbf{u}^s \, \mathrm{d}\Gamma$$

Displacement formulation: weak forms

Solid phase

$$\forall \delta \mathbf{u}^s \in V_s, \quad \int_\Omega \boldsymbol{\sigma}^s(\mathbf{u}^s, \mathbf{u}^f) : \boldsymbol{\epsilon}(\delta \mathbf{u}^s) \, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho}_{11} \mathbf{u}^s \delta \mathbf{u}^s + \widetilde{\rho}_{12} \mathbf{u}^f \delta \mathbf{u}^s \, \mathrm{d}\Omega$$
$$= \oint_{\partial\Omega} \boldsymbol{\sigma}^s(\mathbf{u}^s, \mathbf{u}^f).\mathbf{n} \delta \mathbf{u}^s \, \mathrm{d}\Gamma$$

Fluid phase

$$\forall \delta \mathbf{u}^f \in V_f, \quad \int_\Omega \boldsymbol{\sigma}^f(\mathbf{u}^s, \mathbf{u}^f) : \boldsymbol{\epsilon}(\delta \mathbf{u}^f) \, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho}_{12} \mathbf{u}^s \delta \mathbf{u}^f + \widetilde{\rho}_{22} \mathbf{u}^f \delta \mathbf{u}^f \, \mathrm{d}\Omega$$
$$= \oint_{\partial\Omega} \boldsymbol{\sigma}^f(\mathbf{u}^s, \mathbf{u}^f).\mathbf{n} \delta \mathbf{u}^f \, \mathrm{d}\Gamma$$

Displacement formulation: weak forms

Solid phase

$$\forall \delta \mathbf{u}^s \in V_s, \quad \int_\Omega \boldsymbol{\sigma}^s(\mathbf{u}^s, \mathbf{u}^f) : \boldsymbol{\epsilon}(\delta \mathbf{u}^s)\, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho}_{11} \mathbf{u}^s \delta \mathbf{u}^s + \widetilde{\rho}_{12} \mathbf{u}^f \delta \mathbf{u}^s \, \mathrm{d}\Omega$$
$$= \oint_{\partial\Omega} \boldsymbol{\sigma}^s(\mathbf{u}^s, \mathbf{u}^f).\mathbf{n} \delta \mathbf{u}^s \, \mathrm{d}\Gamma$$

Fluid phase

$$\forall \delta \mathbf{u}^f \in V_f, \quad \int_\Omega \boldsymbol{\sigma}^f(\mathbf{u}^s, \mathbf{u}^f) : \boldsymbol{\epsilon}(\delta \mathbf{u}^f)\, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho}_{12} \mathbf{u}^s \delta \mathbf{u}^f + \widetilde{\rho}_{22} \mathbf{u}^f \delta \mathbf{u}^f \, \mathrm{d}\Omega$$
$$= \oint_{\partial\Omega} \boldsymbol{\sigma}^f(\mathbf{u}^s, \mathbf{u}^f).\mathbf{n} \delta \mathbf{u}^f \, \mathrm{d}\Gamma$$

- ✓ Discretisation of solid and fluid phase
  - ✓ Quadratric for solid phase (shear)
  - ✓ Idem for fluid phase (coupling and Biot waves)
- ✓ Frequency dependent coefficients
  - ✓ Multiplication to “classical” scalar products
  - ✓ Real and sparse matrices
- ✓ Problem of BC

Discretized system

$$\left([\mathcal{K}^s] - \omega^2\,[\mathcal{M}^s]\right)\left\{\begin{array}{c}\mathbf{u}^s\\ \mathbf{u}^f\end{array}\right\} = \left\{\begin{array}{c}\mathbf{F}^s\\ \mathbf{F}^f\end{array}\right\}.$$

6 dofs per node

$$[\mathcal{K}^s] = \left[\begin{array}{cc}\widetilde{A}[\mathbf{K}_0] + N[\mathbf{K}_1] & \widetilde{Q}[\mathbf{K}_0]\\ \widetilde{Q}[\mathbf{K}_0] & \widetilde{R}[\mathbf{K}_0]\end{array}\right]$$

$$[\mathcal{M}^s] = \left[\begin{array}{cc}\widetilde{\rho}_{11}[\mathbf{M}_0] & \widetilde{\rho}_{12}[\mathbf{M}_0]\\ \widetilde{\rho}_{12}[\mathbf{M}_0] & \widetilde{\rho}_{22}[\mathbf{M}_0]\end{array}\right]$$

Discretized system

$$\left([\mathcal{K}^s] - \omega^2 [\mathcal{M}^s]\right) \left\{ \begin{array}{c} \mathbf{u}^s \\ \mathbf{u}^f \end{array} \right\} = \left\{ \begin{array}{c} \mathbf{F}^s \\ \mathbf{F}^f \end{array} \right\}.$$

6 dofs per node

$$[\mathcal{K}^s] = \left[\begin{array}{cc} \widetilde{A}[\mathbf{K}_0] + N[\mathbf{K}_1] & \widetilde{Q}[\mathbf{K}_0] \\ \widetilde{Q}[\mathbf{K}_0] & \widetilde{R}[\mathbf{K}_0] \end{array} \right]$$

$$[\mathcal{M}^s] = \left[\begin{array}{cc} \widetilde{\rho}_{11}[\mathbf{M}_0] & \widetilde{\rho}_{12}[\mathbf{M}_0] \\ \widetilde{\rho}_{12}[\mathbf{M}_0] & \widetilde{\rho}_{22}[\mathbf{M}_0] \end{array} \right]$$

3 sparse real matrices

Discretized system

$$\left([\mathcal{K}^s] - \omega^2 [\mathcal{M}^s]\right) \left\{ \begin{array}{c} \mathbf{u}^s \\ \mathbf{u}^f \end{array} \right\} = \left\{ \begin{array}{c} \mathbf{F}^s \\ \mathbf{F}^f \end{array} \right\}.$$

6 dofs per node

$$[\mathcal{K}^s] = \left[\begin{array}{cc} \widetilde{A}[\mathbf{K}_0] + N[\mathbf{K}_1] & \widetilde{Q}[\mathbf{K}_0] \\ \widetilde{Q}[\mathbf{K}_0] & \widetilde{R}[\mathbf{K}_0] \end{array} \right]$$

$$[\mathcal{M}^s] = \left[\begin{array}{cc} \widetilde{\rho}_{11}[\mathbf{M}_0] & \widetilde{\rho}_{12}[\mathbf{M}_0] \\ \widetilde{\rho}_{12}[\mathbf{M}_0] & \widetilde{\rho}_{22}[\mathbf{M}_0] \end{array} \right]$$

3 sparse real matrices

"Classical form"

- ✓ **Two displacement fields**
- ✓ **Complex system**
- ✓ **Any type of physical model (structural, viscous, thermal)**
- ✓ **Problem of boundary relations**

Total displacement formulation

Total displacement formulation

$$\forall \delta \mathbf{u}^s \in V_s, \quad \int_\Omega \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s) : \boldsymbol{\epsilon}(\delta \mathbf{u}^s)\,\mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho}_s \mathbf{u}^s \delta \mathbf{u}^s + \widetilde{\gamma}\widetilde{\rho}_{eq} \mathbf{u}^f \delta \mathbf{u}^s\,\mathrm{d}\Omega$$

$$= \oint_{\partial\Omega} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta \mathbf{u}^s\,\mathrm{d}\Gamma$$

Total displacement formulation

$$\forall \delta \mathbf{u}^t \in V_f, \quad \int_\Omega -p(\mathbf{u}^t) : \boldsymbol{\epsilon}(\delta \mathbf{u}^t)\, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\gamma}\widetilde{\rho}_{eq} \mathbf{u}^s \delta \mathbf{u}^t + \widetilde{\rho}_{eq} \mathbf{u}^t \delta \mathbf{u}^t \, \mathrm{d}\Omega$$
$$= - \oint_{\partial\Omega} p(\mathbf{u}^t)\mathbf{.n}\delta \mathbf{u}^t \, \mathrm{d}\Gamma$$

Total displacement formulation

$$\forall \delta \mathbf{u}^t \in V_f, \quad \int_\Omega -p(\mathbf{u}^t) : \boldsymbol{\epsilon}(\delta \mathbf{u}^t)\,\mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\gamma}\widetilde{\rho}_{eq}\mathbf{u}^s \delta \mathbf{u}^t + \widetilde{\rho}_{eq}\mathbf{u}^t \delta \mathbf{u}^t\,\mathrm{d}\Omega$$
$$= -\oint_{\partial\Omega} p(\mathbf{u}^t)\boldsymbol{.}\mathbf{n}\delta \mathbf{u}^t\,\mathrm{d}\Gamma$$

$$\left(\left[\mathcal{K}^t\right] - \omega^2\left[\mathcal{M}^t\right]\right)\left\{\begin{array}{c}\mathbf{u}^s\\ \mathbf{u}^f\end{array}\right\} = \left\{\begin{array}{c}\hat{\mathbf{F}}^s\\ \mathbf{F}^t\end{array}\right\}.$$

Total displacement formulation

$$\forall \delta \mathbf{u}^t \in V_f, \quad \int_\Omega -p(\mathbf{u}^t) : \boldsymbol{\epsilon}(\delta \mathbf{u}^t)\, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\gamma}\widetilde{\rho}_{eq} \mathbf{u}^s \delta \mathbf{u}^t + \widetilde{\rho}_{eq} \mathbf{u}^t \delta \mathbf{u}^t \, \mathrm{d}\Omega$$
$$= - \oint_{\partial\Omega} p(\mathbf{u}^t)\mathbf{.n}\delta \mathbf{u}^t \, \mathrm{d}\Gamma$$

$$\left([\mathcal{K}^t] - \omega^2 \, [\mathcal{M}^t] \right) \left\{ \begin{array}{c} \mathbf{u}^s \\ \mathbf{u}^f \end{array} \right\} = \left\{ \begin{array}{c} \hat{\mathbf{F}}^s \\ \mathbf{F}^t \end{array} \right\}.$$

$$[\mathcal{K}^t] = \left[\begin{array}{cc} \hat{A}[\mathbf{K}_0] + N[\mathbf{K}_1] & [\mathbf{0}] \\ [\mathbf{0}] & \widetilde{K}_{eq}[\mathbf{K}_0] \end{array} \right]$$

$$[\mathcal{M}^t] = \left[\begin{array}{cc} \widetilde{\rho}_s[\mathbf{M}_0] & \widetilde{\gamma}\widetilde{\rho}_{eq}[\mathbf{M}_0] \\ \widetilde{\gamma}\widetilde{\rho}_{eq}[\mathbf{M}_0] & \widetilde{\rho}_{eq}[\mathbf{M}_0] \end{array} \right]$$

Total displacement formulation

$$\forall \delta \mathbf{u}^t \in V_f, \quad \int_\Omega -p(\mathbf{u}^t) : \boldsymbol{\epsilon}(\delta \mathbf{u}^t)\,\mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\gamma}\widetilde{\rho}_{eq}\mathbf{u}^s \delta \mathbf{u}^t + \widetilde{\rho}_{eq}\mathbf{u}^t \delta \mathbf{u}^t \,\mathrm{d}\Omega$$
$$= -\oint_{\partial\Omega} p(\mathbf{u}^t).\mathbf{n}\delta \mathbf{u}^t \,\mathrm{d}\Gamma$$

$$\left(\left[\mathcal{K}^t\right] - \omega^2 \left[\mathcal{M}^t\right]\right) \left\{ \begin{array}{c} \mathbf{u}^s \\ \mathbf{u}^f \end{array} \right\} = \left\{ \begin{array}{c} \hat{\mathbf{F}}^s \\ \mathbf{F}^t \end{array} \right\}.$$

$$[\mathcal{K}^t] = \left[\begin{array}{cc} \hat{A}[\mathbf{K}_0] + N[\mathbf{K}_1] & [\mathbf{0}] \\ [\mathbf{0}] & \widetilde{K}_{eq}[\mathbf{K}_0] \end{array} \right]$$

Stress-decoupling

$$[\mathcal{M}^t] = \left[\begin{array}{cc} \widetilde{\rho}_s[\mathbf{M}_0] & \widetilde{\gamma}\widetilde{\rho}_{eq}[\mathbf{M}_0] \\ \widetilde{\gamma}\widetilde{\rho}_{eq}[\mathbf{M}_0] & \widetilde{\rho}_{eq}[\mathbf{M}_0] \end{array} \right]$$

Total displacement formulation

$$\forall \delta \mathbf{u}^t \in V_f, \quad \int_\Omega -p(\mathbf{u}^t) : \boldsymbol{\epsilon}(\delta \mathbf{u}^t)\, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\gamma}\widetilde{\rho}_{eq} \mathbf{u}^s \delta \mathbf{u}^t + \widetilde{\rho}_{eq} \mathbf{u}^t \delta \mathbf{u}^t \, \mathrm{d}\Omega$$
$$= - \oint_{\partial\Omega} p(\mathbf{u}^t).\mathbf{n}\delta \mathbf{u}^t \, \mathrm{d}\Gamma$$

$$\left(\left[\mathcal{K}^t\right] - \omega^2 \left[\mathcal{M}^t\right] \right) \left\{ \begin{array}{c} \mathbf{u}^s \\ \mathbf{u}^f \end{array} \right\} = \left\{ \begin{array}{c} \hat{\mathbf{F}}^s \\ \mathbf{F}^t \end{array} \right\}.$$

$$[\mathcal{K}^t] = \left[\begin{array}{cc} \hat{A}[\mathbf{K}_0] + N[\mathbf{K}_1] & [\mathbf{0}] \\ [\mathbf{0}] & \widetilde{K}_{eq}[\mathbf{K}_0] \end{array} \right]$$

Stress-decoupling

3 sparse real matrices

$$[\mathcal{M}^t] = \left[\begin{array}{cc} \widetilde{\rho}_s[\mathbf{M}_0] & \widetilde{\gamma}\widetilde{\rho}_{eq}[\mathbf{M}_0] \\ \widetilde{\gamma}\widetilde{\rho}_{eq}[\mathbf{M}_0] & \widetilde{\rho}_{eq}[\mathbf{M}_0] \end{array} \right]$$

Interface between 2 PEM

$$\mathbf{u}_1^s = \mathbf{u}_2^s$$

$$\mathbf{u}_1^t.\mathbf{n} = \mathbf{u}_2^t.\mathbf{n}$$

$$p_1 = p_2$$

$$\hat{\boldsymbol{\sigma}}_1.\mathbf{n} = \hat{\boldsymbol{\sigma}}_2.\mathbf{n}$$

Coupling integrals

Interface between 2 PEM

$$\mathbf{u}_1^s = \mathbf{u}_2^s$$
$$\mathbf{u}_1^t.\mathbf{n} = \mathbf{u}_2^t.\mathbf{n}$$
$$p_1 = p_2$$
$$\hat{\boldsymbol{\sigma}}_1.\mathbf{n} = \hat{\boldsymbol{\sigma}}_2.\mathbf{n}$$

$$\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_1^s(\mathbf{u}_1^s).\mathbf{n}\delta\mathbf{u}_1 \,\mathrm{d}\Gamma = \oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_2^s(\mathbf{u}_2^s).\mathbf{n}\delta\mathbf{u}_2 \,\mathrm{d}\Gamma$$
$$\oint_{\partial\Omega_c} p_1(\mathbf{u}^t).\mathbf{n}\delta\mathbf{u}_1 \,\mathrm{d}\Gamma = \oint_{\partial\Omega_c} p_2(\mathbf{u}^t).\mathbf{n}\delta\mathbf{u}_2 \,\mathrm{d}\Gamma$$

Coupling integrals

Interface between 2 PEM

$$
\begin{aligned}
\mathbf{u}_1^s &= \mathbf{u}_2^s \\
\mathbf{u}_1^t.\mathbf{n} &= \mathbf{u}_2^t.\mathbf{n} \\
p_1 &= p_2 \\
\hat{\boldsymbol{\sigma}}_1.\mathbf{n} &= \hat{\boldsymbol{\sigma}}_2.\mathbf{n}
\end{aligned}
$$

$$
\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_1^s(\mathbf{u}_1^s).\mathbf{n}\delta\mathbf{u}_1 \,\mathrm{d}\Gamma = \oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_2^s(\mathbf{u}_2^s).\mathbf{n}\delta\mathbf{u}_2 \,\mathrm{d}\Gamma
$$

$$
\oint_{\partial\Omega_c} p_1(\mathbf{u}^t).\mathbf{n}\delta\mathbf{u}_1 \,\mathrm{d}\Gamma = \oint_{\partial\Omega_c} p_2(\mathbf{u}^t).\mathbf{n}\delta\mathbf{u}_2 \,\mathrm{d}\Gamma
$$

Case of an elastic solid (including septum)

$$
\mathbf{u}^s.\mathbf{n} = \mathbf{u}^t.\mathbf{n} = \mathbf{u}^e.\mathbf{n} \Longrightarrow \delta\mathbf{u}^s.\mathbf{n} = \delta\mathbf{u}^t.\mathbf{n} = \delta\mathbf{u}^e.\mathbf{n}
$$

Coupling integrals

Interface between 2 PEM

$$\begin{aligned} \mathbf{u}_1^s &= \mathbf{u}_2^s \\ \mathbf{u}_1^t.\mathbf{n} &= \mathbf{u}_2^t.\mathbf{n} \\ p_1 &= p_2 \\ \hat{\boldsymbol{\sigma}}_1.\mathbf{n} &= \hat{\boldsymbol{\sigma}}_2.\mathbf{n} \end{aligned}$$

$$\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_1^s(\mathbf{u}_1^s).\mathbf{n}\delta\mathbf{u}_1 \,\mathrm{d}\Gamma = \oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_2^s(\mathbf{u}_2^s).\mathbf{n}\delta\mathbf{u}_2 \,\mathrm{d}\Gamma$$

$$\oint_{\partial\Omega_c} p_1(\mathbf{u}^t).\mathbf{n}\delta\mathbf{u}_1 \,\mathrm{d}\Gamma = \oint_{\partial\Omega_c} p_2(\mathbf{u}^t).\mathbf{n}\delta\mathbf{u}_2 \,\mathrm{d}\Gamma$$

Case of an elastic solid (including septum)

$$\mathbf{u}^s.\mathbf{n} = \mathbf{u}^t.\mathbf{n} = \mathbf{u}^e.\mathbf{n} \implies \delta\mathbf{u}^s.\mathbf{n} = \delta\mathbf{u}^t.\mathbf{n} = \delta\mathbf{u}^e.\mathbf{n}$$

Boundary terms

$$\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u} \,\mathrm{d}\Gamma - \oint_{\partial\Omega_c} p(\mathbf{u}^t).\mathbf{n}\delta\mathbf{u} \,\mathrm{d}\Gamma = \oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}^e(\mathbf{u}^e).\mathbf{n}\delta\mathbf{u} \,\mathrm{d}\Gamma$$

Coupling integrals

Interface between 2 PEM

$$\mathbf{u}_1^s = \mathbf{u}_2^s$$
$$\mathbf{u}_1^t.\mathbf{n} = \mathbf{u}_2^t.\mathbf{n}$$
$$p_1 = p_2$$
$$\hat{\boldsymbol{\sigma}}_1.\mathbf{n} = \hat{\boldsymbol{\sigma}}_2.\mathbf{n}$$

$$\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_1^s(\mathbf{u}_1^s).\mathbf{n}\delta\mathbf{u}_1 \, \mathrm{d}\Gamma = \oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_2^s(\mathbf{u}_2^s).\mathbf{n}\delta\mathbf{u}_2 \, \mathrm{d}\Gamma$$
$$\oint_{\partial\Omega_c} p_1(\mathbf{u}^t).\mathbf{n}\delta\mathbf{u}_1 \, \mathrm{d}\Gamma = \oint_{\partial\Omega_c} p_2(\mathbf{u}^t).\mathbf{n}\delta\mathbf{u}_2 \, \mathrm{d}\Gamma$$

Case of an elastic solid (including septum)

$$\mathbf{u}^s.\mathbf{n} = \mathbf{u}^t.\mathbf{n} = \mathbf{u}^e.\mathbf{n} \Longrightarrow \delta\mathbf{u}^s.\mathbf{n} = \delta\mathbf{u}^t.\mathbf{n} = \delta\mathbf{u}^e.\mathbf{n}$$

Boundary terms

$$\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u} \, \mathrm{d}\Gamma = \oint_{\partial\Omega_c} p(\mathbf{u}^t).\mathbf{n}\delta\mathbf{u} \, \mathrm{d}\Gamma = \oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}^e(\mathbf{u}^e).\mathbf{n}\delta\mathbf{u} \, \mathrm{d}\Gamma$$

Natural coupling: Equality of dof

Porous-air coupling

$$\oint_{\partial\Omega_c} \underbrace{\hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}}_{\mathbf{0}}\,\delta\mathbf{u}^s\,\mathrm{d}\Gamma = 0$$

Porous-air coupling

$$\oint_{\partial\Omega_c} \underbrace{\hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}}_{\mathbf{0}} \, \delta\mathbf{u}^s \, \mathrm{d}\Gamma = 0$$

$$\oint_{\partial\Omega_c} p(\mathbf{u}^t)\delta\mathbf{u}^t.\mathbf{n} \, \mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^a.\mathbf{n}\delta p^a \, \mathrm{d}\Gamma$$

"FS like" term

Porous-air coupling

$$\oint_{\partial\Omega_c} \underbrace{\hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}}_{\mathbf{0}} \, \delta\mathbf{u}^s \, \mathrm{d}\Gamma = 0$$

$$\oint_{\partial\Omega_c} p(\mathbf{u}^t)\delta\mathbf{u}^t.\mathbf{n} \, \mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^a.\mathbf{n}\delta p^a \, \mathrm{d}\Gamma$$

"FS like" term

General form of the linear system

$$\left[\begin{array}{c|cc} \dfrac{[\mathbf{H}_a]}{\rho_a\omega^2} - \dfrac{[\mathbf{Q}_a]}{K_a} & [\mathbf{0}] & [\mathbf{I}_c] \\ \hline [\mathbf{0}] & \hat{A}[\mathbf{K}'] - \omega^2\widetilde{\rho}[\mathbf{M}] & -\omega\widetilde{\gamma}\widetilde{\rho}_{eq}[\mathbf{M}] \\ [\mathbf{I}_c]^t & -\omega\widetilde{\gamma}\widetilde{\rho}_{eq}[\mathbf{M}] & \widetilde{K}_{eq}[\mathbf{K}_0] - \omega\widetilde{\gamma}\widetilde{\rho}_{eq}[\mathbf{M}] \end{array}\right] \left\{\begin{array}{c} \mathbf{P}_a \\ \mathbf{u}^s \\ \mathbf{u}^t \end{array}\right\} = \left\{\begin{array}{c} \mathbf{u}_a \\ \mathbf{F}_s \\ \mathbf{F}_t \end{array}\right\}$$

Convergence

Convergence of poroelastic finite elements based on Biot displacement formulation

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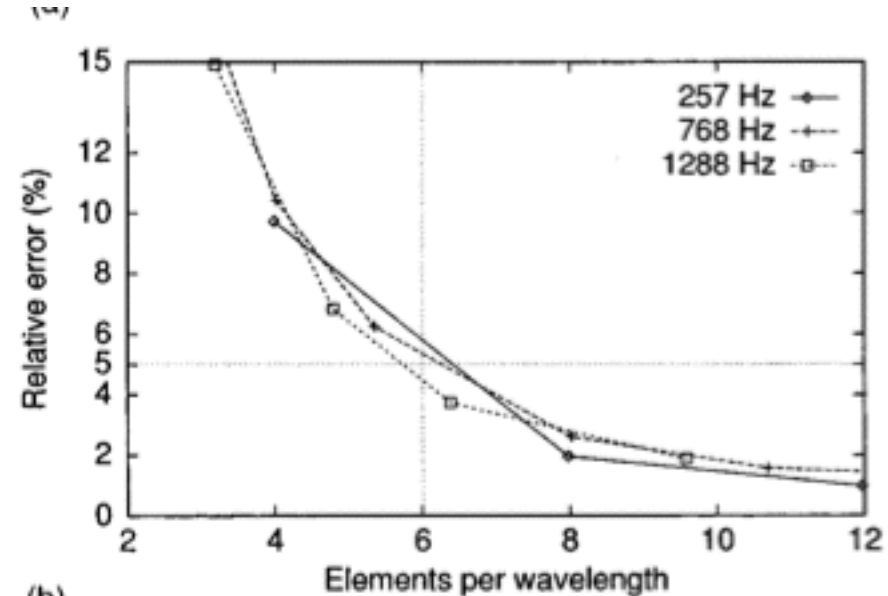
(Received 4 April 1999; revised 15 June 2000; accepted 5 July 2000)

The convergence of linear poroelastic elements based on Biot displacement formulation is investigated. The aim is to determine a mesh criterion that provides reliable results under a given frequency limit. The first part deals with 1D applications for which resonance frequencies can be related to Biot wavelengths. Their relative contributions to the motion are given in order to determine if the mesh criteria for monophasic media are suitable for poroelastic media. The imposition of six linear elements per wavelength is found for each Biot wave as a primary condition for convergence. For 3D applications, convergence rules are derived from a generic configuration, i.e., a clamped porous layer. Because of the complex deformation, the previous criterion is shown to be insufficient. Influence of the coupling between the two phases is demonstrated. © *2001 Acoustical Society of America.* [DOI: 10.1121/1.1289924]

PACS numbers: 43.20.Jr, 43.50.Gf, 43.40.Tm, 43.55.Wk [CBB]



(b)

$\lambda/6$

Convergence

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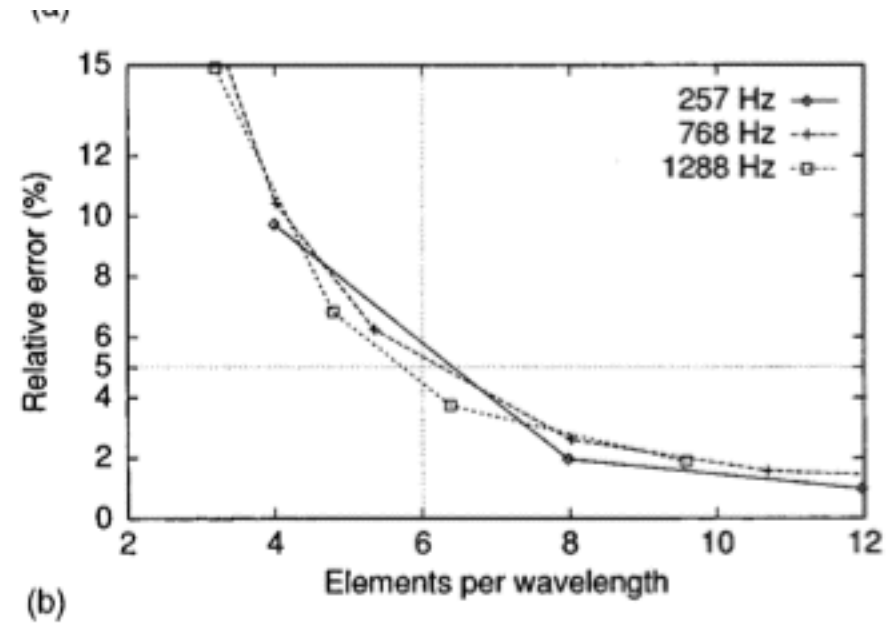

$\lambda/6$

Expressions of dissipated powers and stored energies in poroelastic media modeled by {u,U} and {u,*P*} formulations

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(Received 3 October 2006; revised 10 January 2008; accepted 18 January 2008)

This paper is devoted to the rigorous obtention of the energy balance in porous materials. The wave propagation in the porous media is described by Biot-Allard's {**u**,**U**} and {**u**,*P*} formulations. The paper derives the expressions for stored kinetic and strain energies together with dissipated energies. It is shown that, in the case of mixed formulations, these expressions do not correspond to the real and imaginary parts of the variational formulations. A quantitative convergence analysis of finite element scheme is then undertaken with the help of these indicators. It is shown that the order of convergence of these indicators for linear finite-element is one and that they are then well fitted to check the validity of finite-element models.
 [DOI: 10.1121/1.2874520]

PACS number(s): 43.55.Ev, 43.20.Bi [KA] Pages: 2054–2063

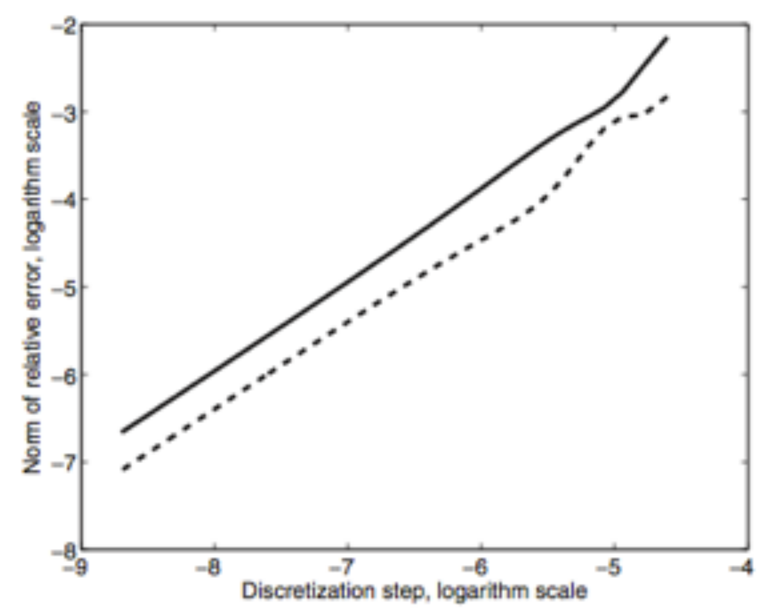
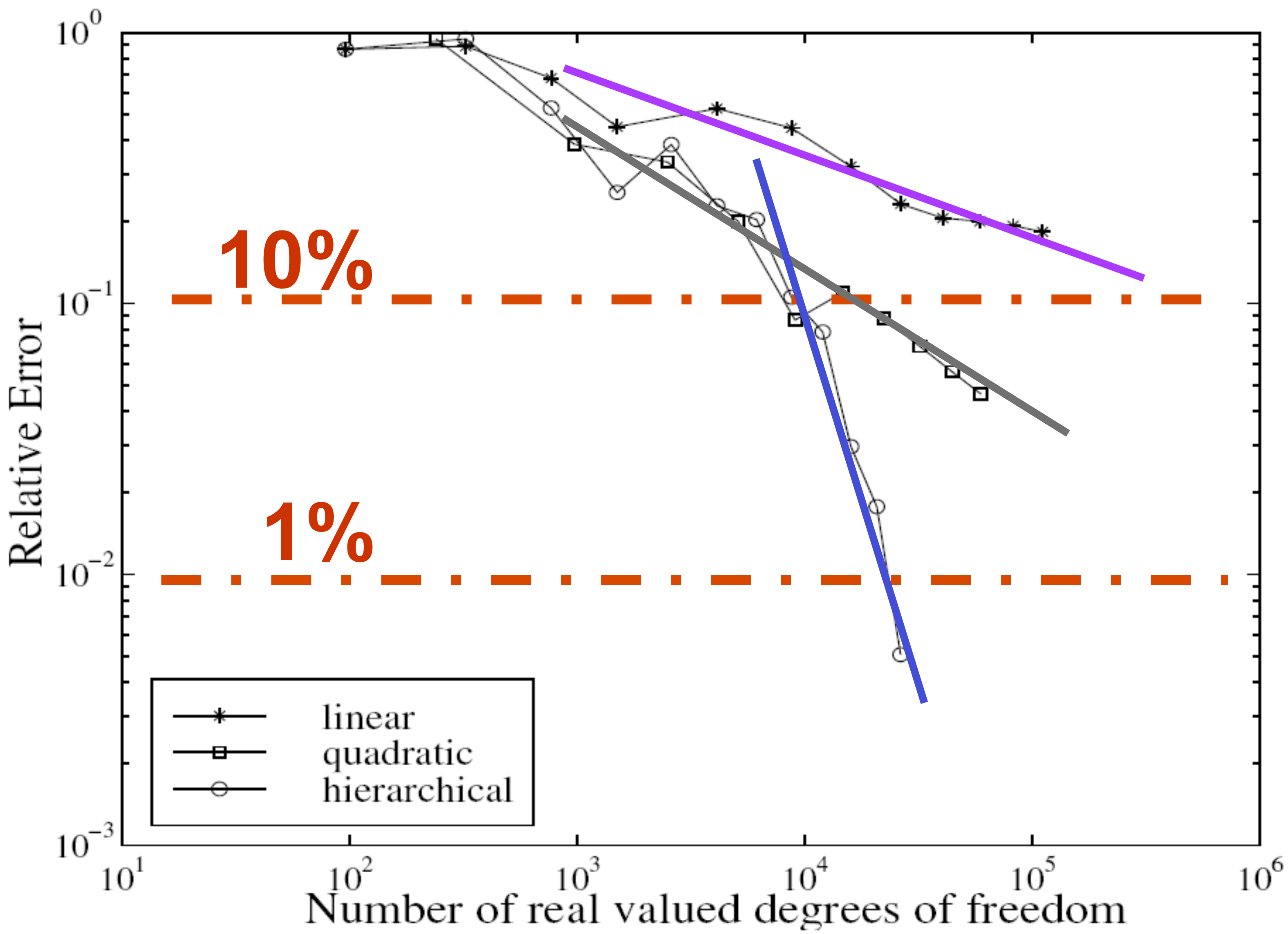


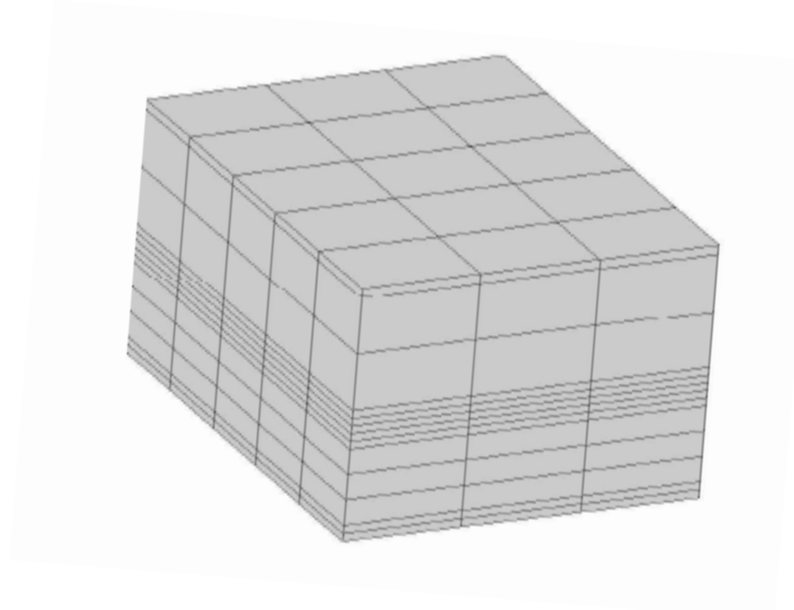
FIG. 6. Convergence of the mean total stored energy. Solid line: {**u**,**U**} formulation and dashed line: {**u**,*P*} formulation.

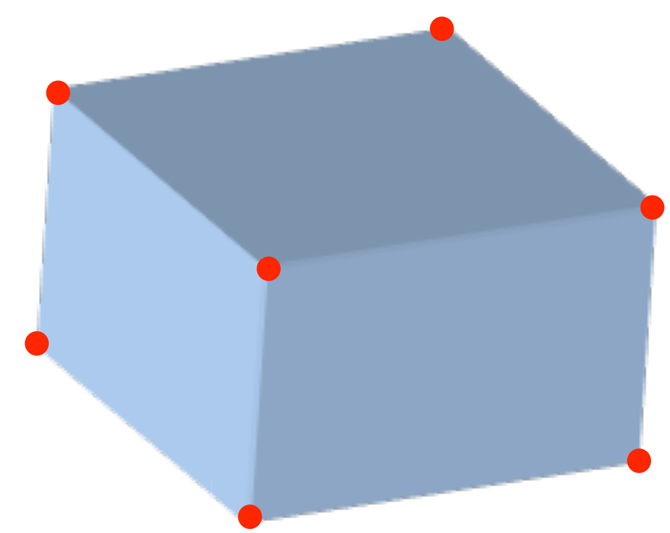
Energetic norms (Hilbert)

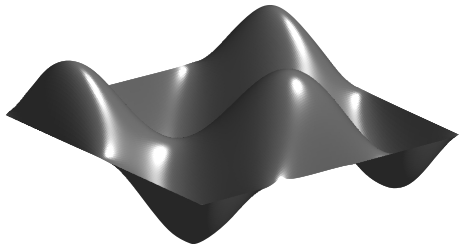
Convergence

Single Block of Foam @ 190Hz



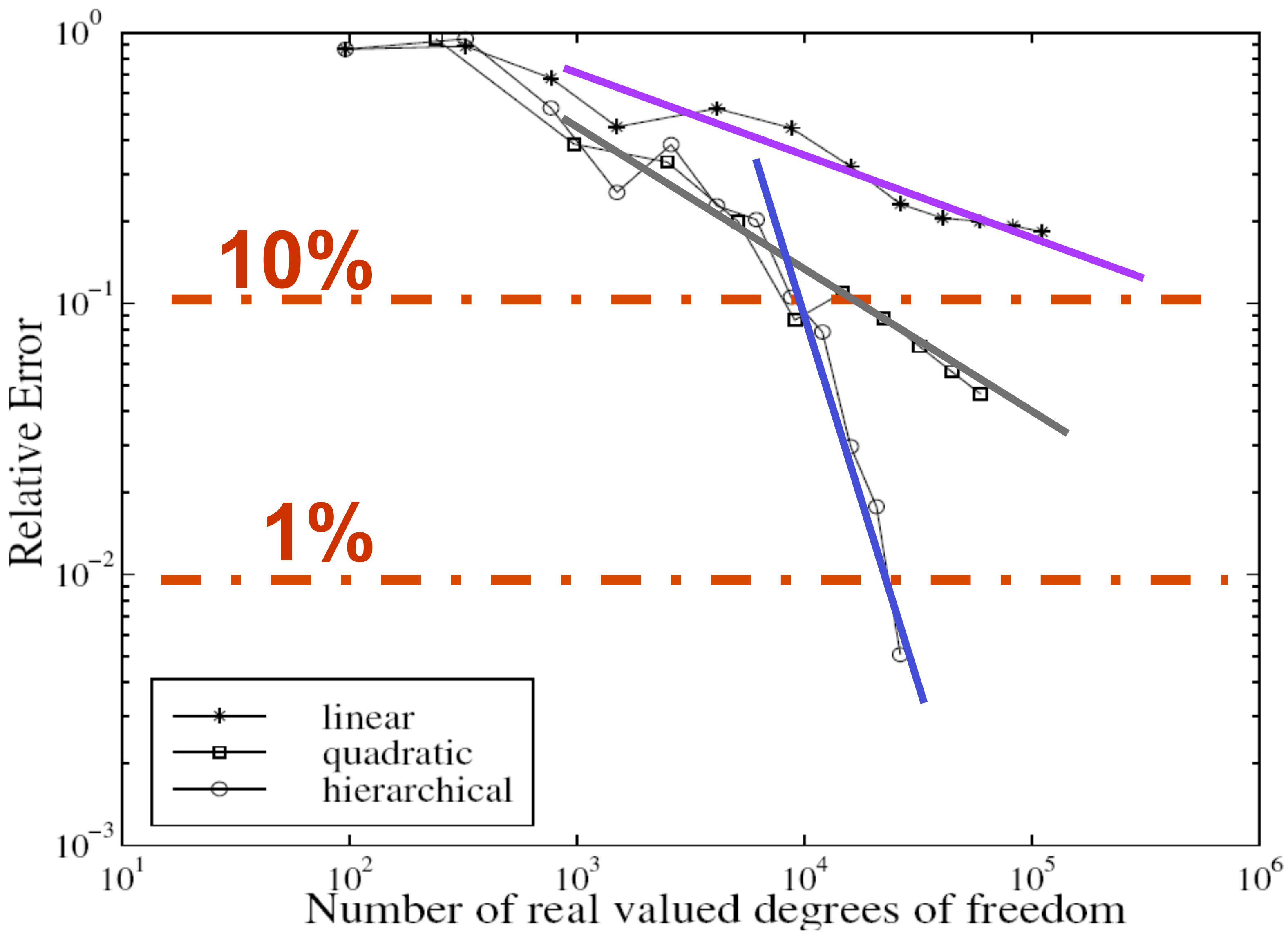


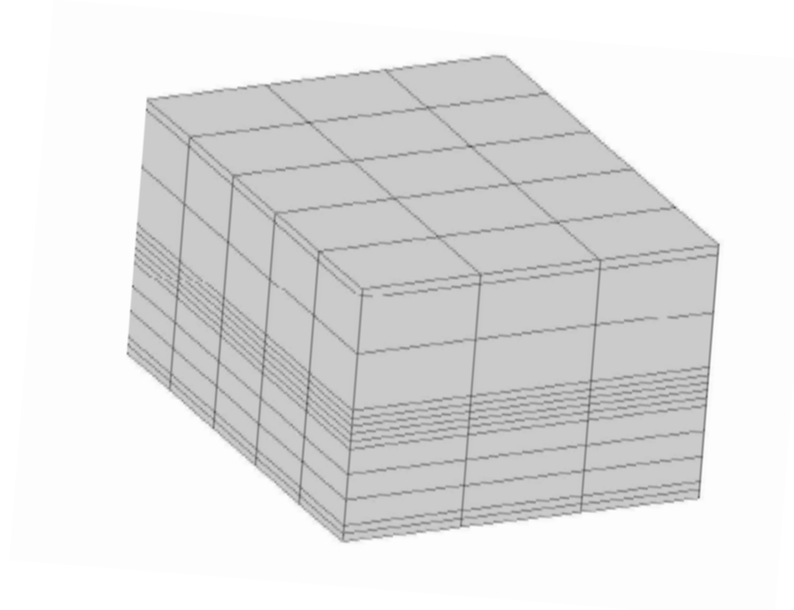


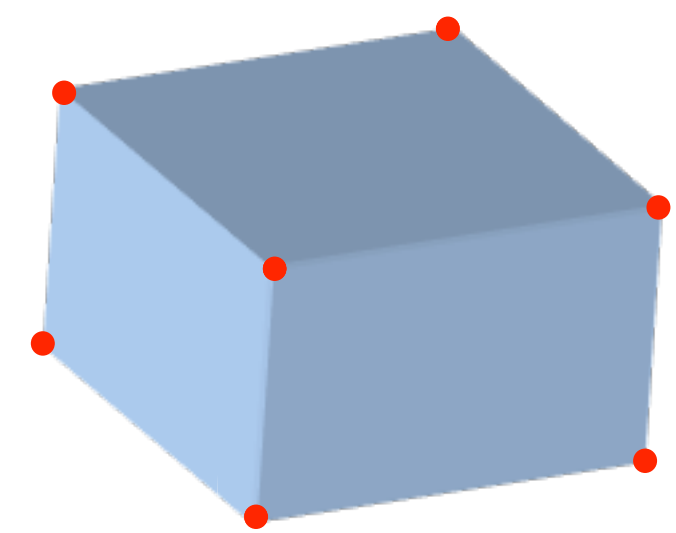


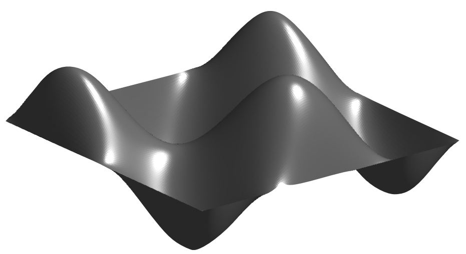
Convergence

Single Block of Foam @ 190Hz









Hörlin *et al.* JSV **245(4)** 2001

Original mixed formulation: weak forms

$$\nabla . \hat{\boldsymbol{\sigma}}(\mathbf{u}^s) = -\omega^2 \widetilde{\rho}\, \mathbf{u}^s - \widetilde{\gamma}\, \nabla\, p \qquad \frac{\nabla^2 p}{\omega^2 \widetilde{\rho}_{eq}} + \frac{p}{\widetilde{K}_{eq}} - \widetilde{\gamma}\, \nabla . \, \mathbf{u}^s = 0$$

Atalla *et al.* JASA **104** 1998

Original mixed formulation: weak forms

$$\nabla.\hat{\boldsymbol{\sigma}}(\mathbf{u}^s) = -\omega^2 \widetilde{\rho}\,\mathbf{u}^s - \widetilde{\gamma}\,\nabla\, p \qquad \frac{\nabla^2 p}{\omega^2 \widetilde{\rho}_{eq}} + \frac{p}{\widetilde{K}_{eq}} - \widetilde{\gamma}\,\nabla.\,\mathbf{u}^s = 0$$

Atalla *et al.* JASA **104** 1998

Solid phase

$$\forall \delta\mathbf{u}^s \in V_s, \quad \int_\Omega \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s) : \boldsymbol{\varepsilon}(\delta\mathbf{u}^s)\,\mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho}\mathbf{u}^s \delta\mathbf{u}^s\,\mathrm{d}\Omega$$
$$- \int_\Omega \widetilde{\gamma}\,\nabla p\,\delta\mathbf{u}^s\,\mathrm{d}\Omega = \oint_{\partial\Omega} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u}^s\,\mathrm{d}\Gamma$$

Original mixed formulation: weak forms

$$\nabla.\hat{\boldsymbol{\sigma}}(\mathbf{u}^s) = -\omega^2\widetilde{\rho}\,\mathbf{u}^s - \widetilde{\gamma}\,\nabla\, p \qquad \frac{\nabla^2 p}{\omega^2\widetilde{\rho}_{eq}} + \frac{p}{\widetilde{K}_{eq}} - \widetilde{\gamma}\,\nabla.\,\mathbf{u}^s = 0$$

Atalla *et al.* JASA **104** 1998

Solid phase

$$\forall \delta\mathbf{u}^s \in V_s, \quad \int_\Omega \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s) : \boldsymbol{\varepsilon}(\delta\mathbf{u}^s)\,\mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho}\mathbf{u}^s\delta\mathbf{u}^s\,\mathrm{d}\Omega$$

$$- \int_\Omega \widetilde{\gamma}\,\nabla p\,\delta\mathbf{u}^s\,\mathrm{d}\Omega = \oint_{\partial\Omega} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u}^s\,\mathrm{d}\Gamma$$

Pressure

$$\forall \delta p \in V_f, \int_\Omega \frac{\boldsymbol{\nabla} p\ \boldsymbol{\nabla}\delta\, p}{\widetilde{\rho}_{eq}\omega^2}\,\mathrm{d}\Omega - \int_\Omega \frac{p\delta p}{\widetilde{K}_{eq}}\,\mathrm{d}\Omega$$

$$- \int_\Omega \widetilde{\gamma}\,\nabla.\mathbf{u}^s\,\delta p\,\mathrm{d}\Omega = \oint_{\partial\Omega} \left[\widetilde{\gamma}\mathbf{u}^s.\mathbf{n} - \frac{1}{\widetilde{\rho}_{eq}}\frac{\partial p}{\partial n}\right]\delta p\,\mathrm{d}\Gamma$$

Orig. mixed form : Discretized system

Atalla *et al.* JASA **104** 1998

$$\begin{bmatrix} \hat{A}[\mathbf{K}] - \omega^2 \widetilde{\rho}[\mathbf{M}] & -\widetilde{\gamma}[\mathbf{C}]^t \\ -\widetilde{\gamma}[\mathbf{C}] & \dfrac{[\mathbf{H}]}{\widetilde{\rho}_{eq}\omega^2} - \dfrac{[\mathbf{Q}]}{\widetilde{K}_{eq}} \end{bmatrix} \left\{ \begin{array}{c} \mathbf{u}^s \\ \mathbf{P} \end{array} \right\} = \left\{ \begin{array}{c} \mathbf{F}_s \\ \mathbf{u}_P \end{array} \right\}$$

Orig. mixed form : Discretized system

Atalla *et al.* JASA **104** 1998

$$\begin{bmatrix} \hat{A}[\mathbf{K}] - \omega^2 \widetilde{\rho}[\mathbf{M}] & -\widetilde{\gamma}[\mathbf{C}]^t \\ -\widetilde{\gamma}[\mathbf{C}] & \dfrac{[\mathbf{H}]}{\widetilde{\rho}_{eq}\omega^2} - \dfrac{[\mathbf{Q}]}{\widetilde{K}_{eq}} \end{bmatrix} \begin{Bmatrix} \mathbf{u}^s \\ \mathbf{P} \end{Bmatrix} = \begin{Bmatrix} \mathbf{F}_s \\ \mathbf{u}_P \end{Bmatrix}$$

5 sparse real matrices

Orig. mixed form : Discretized system

Atalla *et al.* JASA **104** 1998

$$\begin{bmatrix} \hat{A}[\mathbf{K}] - \omega^2 \widetilde{\rho}[\mathbf{M}] & -\widetilde{\gamma}[\mathbf{C}]^t \\ -\widetilde{\gamma}[\mathbf{C}] & \dfrac{[\mathbf{H}]}{\widetilde{\rho}_{eq}\omega^2} - \dfrac{[\mathbf{Q}]}{\widetilde{K}_{eq}} \end{bmatrix} \left\{ \begin{array}{c} \mathbf{u}^s \\ \mathbf{P} \end{array} \right\} = \left\{ \begin{array}{c} \mathbf{F}_s \\ \mathbf{u}_P \end{array} \right\}$$

5 sparse real matrices

"Coupled displacement-pressure system"

✓ 4 dof per node

✓ Complex-frequency dependant

✓ Volumic coupling

✓ Real symmetry

✓ Quite efficient

✓ Mature

Modified mixed formulation: Basic ideas

$$\int_\Omega \nabla p \, \delta \mathbf{u}^s \, \mathrm{d}\Omega = \underbrace{\int_\Omega \nabla \mathbf{.} (p \, \delta \mathbf{u}^s) \, \mathrm{d}\Omega}_{\oint_{\partial\Omega} p \, \delta \mathbf{u}^s \mathbf{.n} \, \mathrm{d}\Gamma} - \int_\Omega p \, \nabla \mathbf{.} \delta \mathbf{u}^s \, \mathrm{d}\Omega$$

Modified mixed formulation: Basic ideas

Atalla *et al.* JASA **109(6)** 2001

$$\int_\Omega \nabla p \, \delta \mathbf{u}^s \, \mathrm{d}\Omega = \underbrace{\int_\Omega \nabla . (p \, \delta \mathbf{u}^s) \, \mathrm{d}\Omega}_{\oint_{\partial\Omega} p \, \delta \mathbf{u}^s . \mathbf{n} \, \mathrm{d}\Gamma} - \int_\Omega p \, \nabla . \delta \mathbf{u}^s \, \mathrm{d}\Omega$$

Modified mixed formulation: Basic ideas

Atalla *et al.* JASA **109(6)** 2001

$$\int_\Omega \nabla p\,\delta\mathbf{u}^s\,\mathrm{d}\Omega = \underbrace{\int_\Omega \nabla\boldsymbol{.}(p\,\delta\mathbf{u}^s)\,\mathrm{d}\Omega}_{\oint_{\partial\Omega} p\,\delta\mathbf{u}^s\boldsymbol{.}\mathbf{n}\;\mathrm{d}\Gamma} - \int_\Omega p\,\nabla\boldsymbol{.}\delta\mathbf{u}^s\,\mathrm{d}\Omega$$

$$\int_\Omega \nabla\boldsymbol{.}\mathbf{u}^s\,\delta p\,\mathrm{d}\Omega = \underbrace{\int_\Omega \nabla\boldsymbol{.}(\delta p\,\mathbf{u}^s)\,\mathrm{d}\Omega}_{\oint_{\partial\Omega} \mathbf{u}^s\boldsymbol{.}\mathbf{n}\,\delta p\;\mathrm{d}\Gamma} - \int_\Omega \mathbf{u}^s\boldsymbol{.}\,\nabla\delta\,p\,\mathrm{d}\Omega$$

Modified mixed formulation: Basic ideas

Atalla *et al.* JASA **109(6)** 2001

$$\int_\Omega \nabla p\,\delta\mathbf{u}^s\,\mathrm{d}\Omega = \underbrace{\int_\Omega \nabla.(p\,\delta\mathbf{u}^s)\,\mathrm{d}\Omega}_{\oint_{\partial\Omega} p\,\delta\mathbf{u}^s.\mathbf{n}\ \mathrm{d}\Gamma} - \int_\Omega p\,\nabla.\delta\mathbf{u}^s\,\mathrm{d}\Omega$$

$$\int_\Omega \nabla.\mathbf{u}^s\,\delta p\,\mathrm{d}\Omega = \underbrace{\int_\Omega \nabla.(\delta p\,\mathbf{u}^s)\,\mathrm{d}\Omega}_{\oint_{\partial\Omega} \mathbf{u}^s.\mathbf{n}\,\delta p\ \mathrm{d}\Gamma} - \int_\Omega \mathbf{u}^s.\nabla\delta\,p\,\mathrm{d}\Omega$$

Interesting relations

$$\widetilde{\gamma} = \frac{\phi}{\widetilde{\alpha}} - 1 \qquad\qquad \phi\left(1 + \frac{\widetilde{Q}}{\widetilde{R}}\right) = 1$$

Modified mixed formulation: Basic ideas

Atalla *et al.* JASA **109(6)** 2001

$$\int_\Omega \nabla p\,\delta\mathbf{u}^s\,\mathrm{d}\Omega = \underbrace{\int_\Omega \nabla.(p\,\delta\mathbf{u}^s)\,\mathrm{d}\Omega}_{\oint_{\partial\Omega} p\,\delta\mathbf{u}^s.\mathbf{n}\;\mathrm{d}\Gamma} - \int_\Omega p\,\nabla.\delta\mathbf{u}^s\,\mathrm{d}\Omega$$

$$\int_\Omega \nabla.\mathbf{u}^s\,\delta p\,\mathrm{d}\Omega = \underbrace{\int_\Omega \nabla.(\delta p\,\mathbf{u}^s)\,\mathrm{d}\Omega}_{\oint_{\partial\Omega} \mathbf{u}^s.\mathbf{n}\,\delta p\;\mathrm{d}\Gamma} - \int_\Omega \mathbf{u}^s.\nabla\delta\,p\,\mathrm{d}\Omega$$

Interesting relations

$$\widetilde{\gamma} = \frac{\phi}{\widetilde{\alpha}} - 1 \qquad\qquad \phi\left(1 + \frac{\widetilde{Q}}{\widetilde{R}}\right) = 1$$

Boundary term

$$\widetilde{\gamma}\mathbf{u}^s.\mathbf{n} - \frac{1}{\widetilde{\rho}_{eq}}\frac{\partial p}{\partial n} = \mathbf{u}^t.\mathbf{n}$$

Modified mixed formulation: solid phase

Original formulation

$$\forall \delta \mathbf{u}^s \in V_s, \quad \int_\Omega \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s) : \boldsymbol{\varepsilon}(\delta \mathbf{u}^s)\, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho} \mathbf{u}^s \delta \mathbf{u}^s \, \mathrm{d}\Omega$$

$$- \int_\Omega \widetilde{\gamma}\, \nabla p\, \delta \mathbf{u}^s \, \mathrm{d}\Omega = \oint_{\partial\Omega} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n} \delta \mathbf{u}^s \, \mathrm{d}\Gamma$$

Modified mixed formulation: solid phase

Original formulation

$$\forall \delta \mathbf{u}^s \in V_s, \quad \int_\Omega \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s) : \boldsymbol{\varepsilon}(\delta \mathbf{u}^s)\,\mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho}\mathbf{u}^s \delta \mathbf{u}^s\,\mathrm{d}\Omega$$

$$- \int_\Omega \widetilde{\gamma}\,\nabla p\,\delta \mathbf{u}^s\,\mathrm{d}\Omega = \oint_{\partial\Omega} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta \mathbf{u}^s\,\mathrm{d}\Gamma$$

Rewriting

$$\widetilde{\gamma} \int_\Omega \nabla p\,\delta \mathbf{u}^s\,\mathrm{d}\Omega - \oint_{\partial\Omega} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta \mathbf{u}^s\,\mathrm{d}\Gamma$$

$$= \frac{1}{\widetilde{\alpha}} \int_\Omega \nabla p\,\delta \mathbf{u}^s\,\mathrm{d}\Omega - \int_\Omega p\,\nabla.\delta \mathbf{u}^s\,\mathrm{d}\Omega - \oint_{\partial\Omega} \underbrace{(\hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s) - p\boldsymbol{\delta})}_{\boldsymbol{\sigma}^t(\mathbf{u}^s, p)}.\mathbf{n}\delta \mathbf{u}^s\,\mathrm{d}\Gamma$$

Modified mixed formulation: solid phase

Original formulation

$$\forall \delta \mathbf{u}^s \in V_s, \quad \int_\Omega \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s) : \boldsymbol{\varepsilon}(\delta \mathbf{u}^s)\, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho} \mathbf{u}^s \delta \mathbf{u}^s \, \mathrm{d}\Omega$$

$$- \int_\Omega \widetilde{\gamma}\, \nabla p\, \delta \mathbf{u}^s \, \mathrm{d}\Omega = \oint_{\partial\Omega} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n} \delta \mathbf{u}^s \, \mathrm{d}\Gamma$$

Rewriting

$$\widetilde{\gamma} \int_\Omega \nabla p\, \delta \mathbf{u}^s \, \mathrm{d}\Omega - \oint_{\partial\Omega} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n} \delta \mathbf{u}^s \, \mathrm{d}\Gamma$$

$$= \frac{1}{\widetilde{\alpha}} \int_\Omega \nabla p\, \delta \mathbf{u}^s \, \mathrm{d}\Omega - \int_\Omega p\, \nabla . \delta \mathbf{u}^s \, \mathrm{d}\Omega - \oint_{\partial\Omega} \underbrace{(\hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s) - p \boldsymbol{\delta})}_{\boldsymbol{\sigma}^t(\mathbf{u}^s, p)} .\mathbf{n} \delta \mathbf{u}^s \, \mathrm{d}\Gamma$$

Modified formulation

$$\forall \delta \mathbf{u}^s \in V_s, \quad \int_\Omega \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s) : \boldsymbol{\varepsilon}(\delta \mathbf{u}^s)\, \mathrm{d}\Omega - \omega^2 \int_\Omega \widetilde{\rho} \mathbf{u}^s \delta \mathbf{u}^s \, \mathrm{d}\Omega$$

$$- \int_\Omega \frac{\phi}{\widetilde{\alpha}} \nabla p\, \delta \mathbf{u}^s \, \mathrm{d}\Omega - \int_\Omega p\, \nabla . \delta \mathbf{u}^s \, \mathrm{d}\Omega = \oint_{\partial\Omega} \boldsymbol{\sigma}^t(\mathbf{u}^s, p).\mathbf{n} \delta \mathbf{u}^s \, \mathrm{d}\Gamma$$

Modified mixed formulation: pressure

Original formulation

$$\forall \delta p \in V_f, \int_\Omega \frac{\boldsymbol{\nabla} p \; \boldsymbol{\nabla} \delta\, p}{\widetilde{\rho}_{eq}\omega^2} \, \mathrm{d}\Omega - \int_\Omega \frac{p\delta p}{\widetilde{K}_{eq}} \, \mathrm{d}\Omega$$

$$- \int_\Omega \widetilde{\gamma}\, \nabla.\mathbf{u}^s \, \delta p \, \mathrm{d}\Omega = \oint_{\partial\Omega} \mathbf{u}^t.\mathbf{n}\delta p \, \mathrm{d}\Gamma$$

Original formulation

$$\forall \delta p \in V_f, \int_\Omega \frac{\boldsymbol{\nabla} p \; \boldsymbol{\nabla} \delta p}{\widetilde{\rho}_{eq}\omega^2} \, \mathrm{d}\Omega - \int_\Omega \frac{p \delta p}{\widetilde{K}_{eq}} \, \mathrm{d}\Omega$$

$$- \int_\Omega \widetilde{\gamma} \, \nabla . \mathbf{u}^s \, \delta p \, \mathrm{d}\Omega = \oint_{\partial\Omega} \mathbf{u}^t . \mathbf{n} \delta p \, \mathrm{d}\Gamma$$

Rewriting

$$\widetilde{\gamma} \int_\Omega \nabla . \mathbf{u}^s \, \delta p \, \mathrm{d}\Omega - \oint_{\partial\Omega} \mathbf{u}^t . \mathbf{n} \delta p \, \mathrm{d}\Gamma$$

$$= \frac{1}{\widetilde{\alpha}} \int_\Omega \nabla . \mathbf{u}^s \, \delta p \, \mathrm{d}\Omega - \int_\Omega \mathbf{u}^s . \nabla \delta p \, \mathrm{d}\Omega - \oint_{\partial\Omega} \underbrace{(\mathbf{u}^t - \mathbf{u}^s)}_{\mathbf{w} = \phi(\mathbf{u}^f - \mathbf{u}^s)} . \mathbf{n} \delta p \, \mathrm{d}\Gamma$$

Original formulation

$$\forall \delta p \in V_f, \int_\Omega \frac{\boldsymbol{\nabla} p \; \boldsymbol{\nabla} \delta \, p}{\widetilde{\rho}_{eq} \omega^2} \, \mathrm{d}\Omega - \int_\Omega \frac{p \delta p}{\widetilde{K}_{eq}} \, \mathrm{d}\Omega$$

$$- \int_\Omega \widetilde{\gamma} \, \nabla . \mathbf{u}^s \, \delta p \, \mathrm{d}\Omega = \oint_{\partial\Omega} \mathbf{u}^t . \mathbf{n} \delta p \, \mathrm{d}\Gamma$$

Rewriting

$$\widetilde{\gamma} \int_\Omega \nabla . \mathbf{u}^s \, \delta p \, \mathrm{d}\Omega - \oint_{\partial\Omega} \mathbf{u}^t . \mathbf{n} \delta p \, \mathrm{d}\Gamma$$

$$= \frac{1}{\widetilde{\alpha}} \int_\Omega \nabla . \mathbf{u}^s \, \delta p \, \mathrm{d}\Omega - \int_\Omega \mathbf{u}^s . \nabla \delta \, p \, \mathrm{d}\Omega - \oint_{\partial\Omega} \underbrace{(\mathbf{u}^t - \mathbf{u}^s)}_{\mathbf{w} = \phi(\mathbf{u}^f - \mathbf{u}^s)} . \mathbf{n} \delta p \, \mathrm{d}\Gamma$$

Modified formulation

$$\forall \delta p \in V_f, \int_\Omega \frac{\boldsymbol{\nabla} p \; \boldsymbol{\nabla} \delta \, p}{\widetilde{\rho}_{eq} \omega^2} \, \mathrm{d}\Omega - \int_\Omega \frac{p \delta p}{\widetilde{K}_{eq}} \, \mathrm{d}\Omega$$

$$- \frac{\phi}{\widetilde{\alpha}} \int_\Omega \nabla . \mathbf{u}^s \, \delta p \, \mathrm{d}\Omega - \int_\Omega \mathbf{u}^s \; \nabla \delta \, p \, \mathrm{d}\Omega - = \oint_{\partial\Omega} \mathbf{w} . \mathbf{n} \delta p \, \mathrm{d}\Gamma$$

Modified mixed formulation: system

$$\begin{bmatrix} \hat{A}[\mathbf{K}] - \omega^2\widetilde{\rho}[\mathbf{M}] & -\dfrac{\phi}{\widetilde{\alpha}}[\mathbf{C}]^t - [\mathbf{C}']^t \\ -\dfrac{\phi}{\widetilde{\alpha}}\widetilde{\gamma}[\mathbf{C}] - [\mathbf{C}'] & \dfrac{[\mathbf{H}]}{\widetilde{\rho}_{eq}\omega^2} - \dfrac{[\mathbf{Q}]}{\widetilde{K}_{eq}} \end{bmatrix} \begin{Bmatrix} \mathbf{u}^s \\ \mathbf{P} \end{Bmatrix} = \begin{Bmatrix} \mathbf{F}_t \\ \mathbf{u}_\mathbf{w} \end{Bmatrix}$$

Modified mixed formulation: system

$$\begin{bmatrix} \hat{A}[\mathbf{K}] - \omega^2 \widetilde{\rho}[\mathbf{M}] & -\dfrac{\phi}{\widetilde{\alpha}}[\mathbf{C}]^t - [\mathbf{C}']^t \\ -\dfrac{\phi}{\widetilde{\alpha}}\widetilde{\gamma}[\mathbf{C}] - [\mathbf{C}'] & \dfrac{[\mathbf{H}]}{\widetilde{\rho}_{eq}\omega^2} - \dfrac{[\mathbf{Q}]}{\widetilde{K}_{eq}} \end{bmatrix} \begin{Bmatrix} \mathbf{u}^s \\ \mathbf{P} \end{Bmatrix} = \begin{Bmatrix} \mathbf{F}_t \\ \mathbf{u_w} \end{Bmatrix}$$

6 sparse real matrices

Modified mixed formulation: system

$$\begin{bmatrix} \hat{A}[\mathbf{K}] - \omega^2 \widetilde{\rho}[\mathbf{M}] & -\frac{\phi}{\widetilde{\alpha}}[\mathbf{C}]^t - [\mathbf{C}']^t \\ -\frac{\phi}{\widetilde{\alpha}}\widetilde{\gamma}[\mathbf{C}] - [\mathbf{C}'] & \frac{[\mathbf{H}]}{\widetilde{\rho}_{eq}\omega^2} - \frac{[\mathbf{Q}]}{\widetilde{K}_{eq}} \end{bmatrix} \begin{Bmatrix} \mathbf{u}^s \\ \mathbf{P} \end{Bmatrix} = \begin{Bmatrix} \mathbf{F}_t \\ \mathbf{u_w} \end{Bmatrix}$$

Atalla *et al.* JASA **109(6)** 2001

6 sparse real matrices

"Coupled displacement-pressure system"

✓ One more volumic coupling matrice

✓ Different surfacic integrals

- **✓ Total stress**
- **✓ Relative flow**

Interface between 2 PEM

$$\mathbf{u}_1^s = \mathbf{u}_2^s$$

$$\mathbf{u}_1^t.\mathbf{n} = \mathbf{u}_2^t.\mathbf{n}$$

$$p_1 = p_2$$

$$\hat{\boldsymbol{\sigma}}_1^s.\mathbf{n} = \hat{\boldsymbol{\sigma}}_2^s.\mathbf{n}$$

$$\boldsymbol{\sigma}_1^t.\mathbf{n} = \boldsymbol{\sigma}_2^t.\mathbf{n}$$

uP original

$$\mathbf{u}_1^s = \mathbf{u}_2^s$$

$$\mathbf{u}_1^t.\mathbf{n} = \mathbf{u}_2^t.\mathbf{n}$$

$$p_1 = p_2$$

$$\hat{\boldsymbol{\sigma}}_1^s.\mathbf{n} = \hat{\boldsymbol{\sigma}}_2^s.\mathbf{n}$$

$$\boldsymbol{\sigma}_1^t.\mathbf{n} = \boldsymbol{\sigma}_2^t.\mathbf{n}$$

$$\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_1^s(\mathbf{u}_1^s).\mathbf{n}\delta\mathbf{u}_1 \,\mathrm{d}\Gamma = \oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_2^s(\mathbf{u}_2^s).\mathbf{n}\delta\mathbf{u}_2 \,\mathrm{d}\Gamma$$

$$\oint_{\partial\Omega} \mathbf{u}_1^t.\mathbf{n}\delta p \,\mathrm{d}\Gamma = \oint_{\partial\Omega} \mathbf{u}_2^t.\mathbf{n}\delta p \,\mathrm{d}\Gamma$$

$$\mathbf{u}_1^s = \mathbf{u}_2^s$$

$$\mathbf{u}_1^t.\mathbf{n} = \mathbf{u}_2^t.\mathbf{n}$$

$$p_1 = p_2$$

$$\hat{\boldsymbol{\sigma}}_1^s.\mathbf{n} = \hat{\boldsymbol{\sigma}}_2^s.\mathbf{n}$$

$$\boldsymbol{\sigma}_1^t.\mathbf{n} = \boldsymbol{\sigma}_2^t.\mathbf{n}$$

uP original

$$\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_1^s(\mathbf{u}_1^s).\mathbf{n}\delta\mathbf{u}_1 \, \mathrm{d}\Gamma = \oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_2^s(\mathbf{u}_2^s).\mathbf{n}\delta\mathbf{u}_2 \, \mathrm{d}\Gamma$$

$$\oint_{\partial\Omega} \mathbf{u}_1^t.\mathbf{n}\delta p \, \mathrm{d}\Gamma = \oint_{\partial\Omega} \mathbf{u}_2^t.\mathbf{n}\delta p \, \mathrm{d}\Gamma$$

uP modified

$$\oint_{\partial\Omega_c} \boldsymbol{\sigma}_1^t.\mathbf{n}\delta\mathbf{u}_1 \, \mathrm{d}\Gamma = \oint_{\partial\Omega_c} \boldsymbol{\sigma}_2^t.\mathbf{n}\delta\mathbf{u}_2 \, \mathrm{d}\Gamma$$

$$\oint_{\partial\Omega} \mathbf{w}_1.\mathbf{n}\delta p \, \mathrm{d}\Gamma = \oint_{\partial\Omega} \mathbf{w}_2.\mathbf{n}\delta p \, \mathrm{d}\Gamma$$

$$
\begin{aligned}
\mathbf{u}_1^s &= \mathbf{u}_2^s \\
\mathbf{u}_1^t.\mathbf{n} &= \mathbf{u}_2^t.\mathbf{n} \\
p_1 &= p_2 \\
\hat{\boldsymbol{\sigma}}_1^s.\mathbf{n} &= \hat{\boldsymbol{\sigma}}_2^s.\mathbf{n} \\
\boldsymbol{\sigma}_1^t.\mathbf{n} &= \boldsymbol{\sigma}_2^t.\mathbf{n}
\end{aligned}
$$

uP original

$$\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_1^s(\mathbf{u}_1^s).\mathbf{n}\delta\mathbf{u}_1 \, \mathrm{d}\Gamma = \oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}_2^s(\mathbf{u}_2^s).\mathbf{n}\delta\mathbf{u}_2 \, \mathrm{d}\Gamma$$

$$\oint_{\partial\Omega} \mathbf{u}_1^t.\mathbf{n}\delta p \, \mathrm{d}\Gamma = \oint_{\partial\Omega} \mathbf{u}_2^t.\mathbf{n}\delta p \, \mathrm{d}\Gamma$$

uP modified

$$\oint_{\partial\Omega_c} \boldsymbol{\sigma}_1^t.\mathbf{n}\delta\mathbf{u}_1 \, \mathrm{d}\Gamma = \oint_{\partial\Omega_c} \boldsymbol{\sigma}_2^t.\mathbf{n}\delta\mathbf{u}_2 \, \mathrm{d}\Gamma$$

$$\oint_{\partial\Omega} \mathbf{w}_1.\mathbf{n}\delta p \, \mathrm{d}\Gamma = \oint_{\partial\Omega} \mathbf{w}_2.\mathbf{n}\delta p \, \mathrm{d}\Gamma$$

Natural coupling

PEM - elastic solid interface

$$\mathbf{u}^s = \mathbf{u}^e$$

$$\mathbf{u}^s.\mathbf{n} = \mathbf{u}^t.\mathbf{n}$$

$$\left(\hat{\boldsymbol{\sigma}} - p\boldsymbol{\delta}\right).\mathbf{n} = \boldsymbol{\sigma}^e.\mathbf{n}$$

PEM - elastic solid interface

$$\mathbf{u}^s = \mathbf{u}^e$$

$$\mathbf{u}^s.\mathbf{n} = \mathbf{u}^t.\mathbf{n}$$

$$\left(\hat{\boldsymbol{\sigma}} - p\boldsymbol{\delta}\right).\mathbf{n} = \boldsymbol{\sigma}^e.\mathbf{n}$$

Original formulation

$$\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u}^s \,\mathrm{d}\Gamma - \oint_{\partial\Omega_c} \boldsymbol{\sigma}^e(\mathbf{u}^e).\mathbf{n}\delta\mathbf{u}^e \,\mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^t.\mathbf{n}\delta p \,\mathrm{d}\Gamma$$

PEM - elastic solid interface

$$\mathbf{u}^s = \mathbf{u}^e$$

$$\mathbf{u}^s.\mathbf{n} = \mathbf{u}^t.\mathbf{n}$$

$$\left(\hat{\boldsymbol{\sigma}} - p\boldsymbol{\delta}\right).\mathbf{n} = \boldsymbol{\sigma}^e.\mathbf{n}$$

Original formulation

$$\oint_{\partial\Omega_c} p\delta\mathbf{u}^s.\mathbf{n}\,\mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^t.\mathbf{n}\delta p\,\mathrm{d}\Gamma$$

PEM - elastic solid interface

$$\mathbf{u}^s = \mathbf{u}^e$$

$$\mathbf{u}^s.\mathbf{n} = \mathbf{u}^t.\mathbf{n}$$

$$\left(\hat{\boldsymbol{\sigma}} - p\boldsymbol{\delta}\right).\mathbf{n} = \boldsymbol{\sigma}^e.\mathbf{n}$$

Original formulation

$$\oint_{\partial\Omega_c} p\delta\mathbf{u}^s.\mathbf{n}\,\mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^t.\mathbf{n}\delta p\,\mathrm{d}\Gamma$$

Modified formulation

$$\oint_{\partial\Omega_c} \boldsymbol{\sigma}^t(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u}^s\,\mathrm{d}\Gamma - \oint_{\partial\Omega_c} \boldsymbol{\sigma}^e(\mathbf{u}^e).\mathbf{n}\delta\mathbf{u}^e\,\mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{w}^t.\mathbf{n}\delta p\,\mathrm{d}\Gamma$$

PEM - elastic solid interface

$$\mathbf{u}^s = \mathbf{u}^e$$

$$\mathbf{u}^s.\mathbf{n} = \mathbf{u}^t.\mathbf{n}$$

$$\left(\hat{\boldsymbol{\sigma}} - p\boldsymbol{\delta}\right).\mathbf{n} = \boldsymbol{\sigma}^e.\mathbf{n}$$

Original formulation

$$\oint_{\partial\Omega_c} p\delta\mathbf{u}^s.\mathbf{n}\,\mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^t.\mathbf{n}\delta p\,\mathrm{d}\Gamma$$

Modified formulation

$$\oint_{\partial\Omega_c} \boldsymbol{\sigma}^t(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u}^s\,\mathrm{d}\Gamma = \oint_{\partial\Omega_c} \boldsymbol{\sigma}^e(\mathbf{u}^e).\mathbf{n}\delta\mathbf{u}^e\,\mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{w}^t.\mathbf{n}\delta p\,\mathrm{d}\Gamma$$

PEM - air interface

$$\mathbf{u}^t.\mathbf{n} = -\frac{1}{\rho_a \omega^2}\frac{\partial p^a}{\partial n}$$

$$p = p^a$$

$$\hat{\boldsymbol{\sigma}}.\mathbf{n} = \mathbf{0}$$

PEM - air interface

$$\mathbf{u}^t.\mathbf{n} = -\frac{1}{\rho_a \omega^2}\frac{\partial p^a}{\partial n}$$

$$p = p^a$$

$$\hat{\boldsymbol{\sigma}}.\mathbf{n} = \mathbf{0}$$

Original formulation

$$\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u}^s \,\mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^t.\mathbf{n}\delta p \,\mathrm{d}\Gamma - \oint_{\partial\Omega_c} \mathbf{u}^a.\mathbf{n}\delta a \,\mathrm{d}\Gamma$$

PEM - air interface

$$\mathbf{u}^t.\mathbf{n} = -\frac{1}{\rho_a\omega^2}\frac{\partial p^a}{\partial n}$$

$$p = p^a$$

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Original formulation

$$\xcancel{\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u}^s \,\mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^t.\mathbf{n}\delta p \,\mathrm{d}\Gamma - \oint_{\partial\Omega_c} \mathbf{u}^a.\mathbf{n}\delta a \,\mathrm{d}\Gamma}$$

PEM - air interface

$$\mathbf{u}^t.\mathbf{n} = -\frac{1}{\rho_a \omega^2}\frac{\partial p^a}{\partial n}$$

$$p = p^a$$

$$\hat{\boldsymbol{\sigma}}.\mathbf{n} = \mathbf{0}$$

Original formulation

$$\xcancel{\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u}^s \, d\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^t.\mathbf{n}\delta p \, d\Gamma - \oint_{\partial\Omega_c} \mathbf{u}^a.\mathbf{n}\delta a \, d\Gamma}$$

Modified formulation

$$\oint_{\partial\Omega_c} \boldsymbol{\sigma}^t(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u}^s \, d\Gamma + \oint_{\partial\Omega_c} \mathbf{w}^t.\mathbf{n}\delta p \, d\Gamma - \oint_{\partial\Omega_c} \mathbf{u}^a.\mathbf{n}\delta p^a \, d\Gamma$$

PEM - air interface

$$\mathbf{u}^t.\mathbf{n} = -\frac{1}{\rho_a \omega^2}\frac{\partial p^a}{\partial n}$$

$$p = p^a$$

$$\hat{\boldsymbol{\sigma}}.\mathbf{n} = \mathbf{0}$$

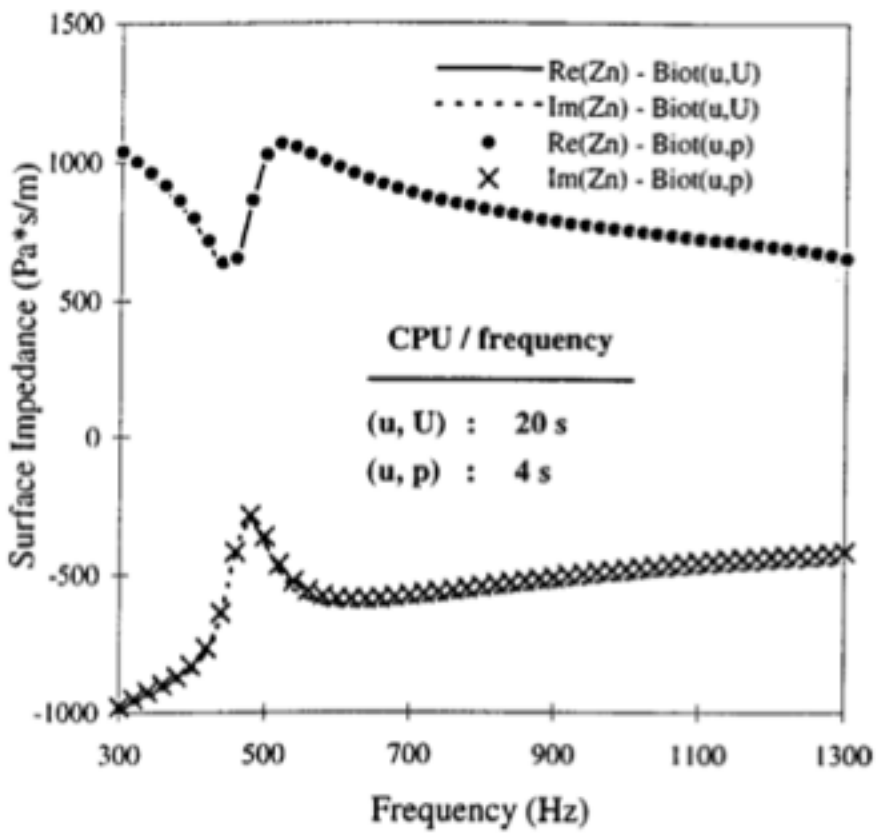
Original formulation

$$\xcancel{\oint_{\partial\Omega_c} \hat{\boldsymbol{\sigma}}^s(\mathbf{u}^s).\mathbf{n}\delta\mathbf{u}^s \, \mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^t.\mathbf{n}\delta p \, \mathrm{d}\Gamma - \oint_{\partial\Omega_c} \mathbf{u}^a.\mathbf{n}\delta a \, \mathrm{d}\Gamma}$$

Modified formulation

$$\oint_{\partial\Omega_c} p\delta\mathbf{u}^s.\mathbf{n} \, \mathrm{d}\Gamma + \oint_{\partial\Omega_c} \mathbf{u}^s.\mathbf{n}\delta p \, \mathrm{d}\Gamma$$

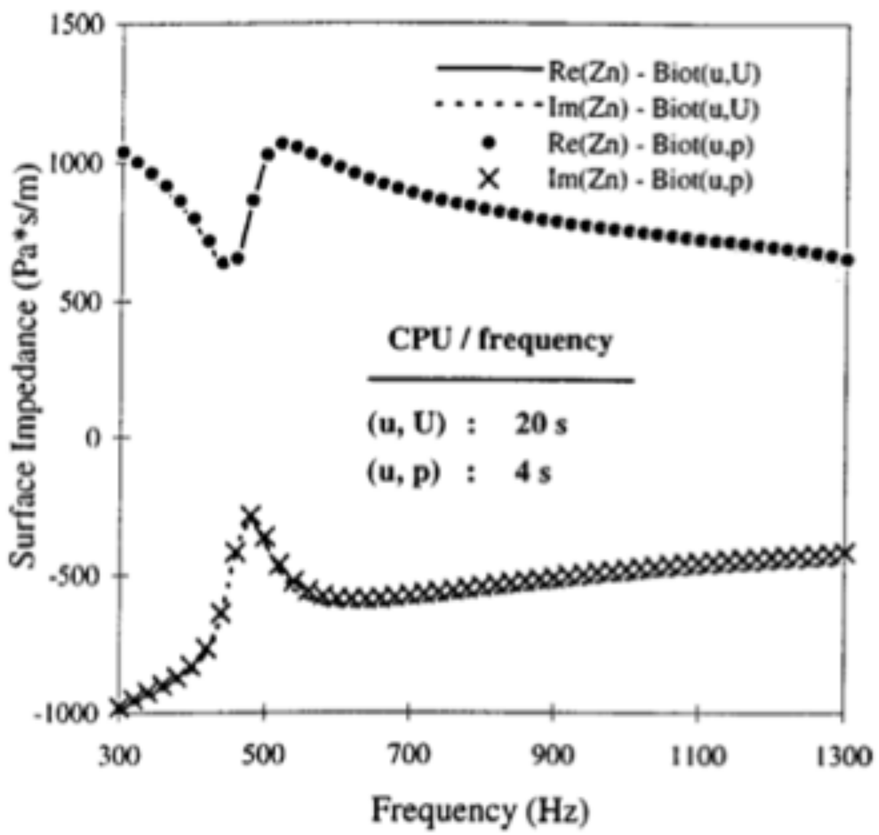
Mixed formulation: results



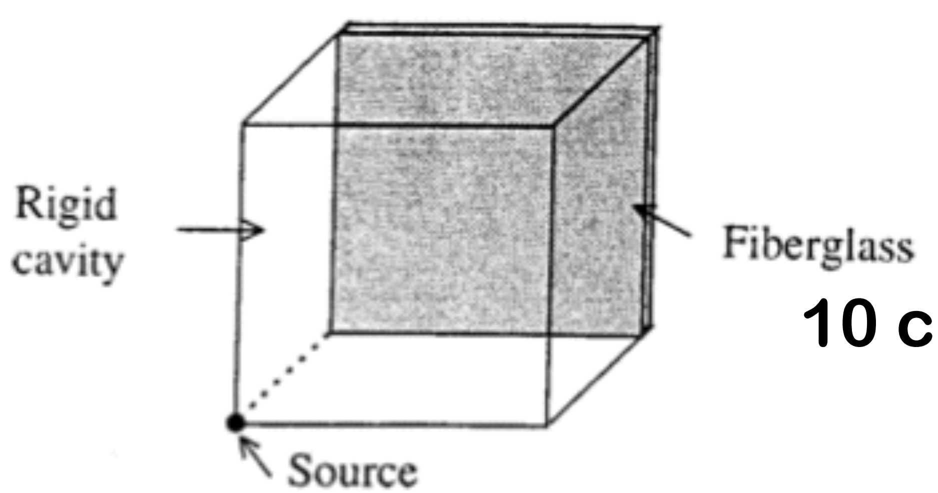
Atalla *et al.* JASA **104** 1998

Mixed formulation: results

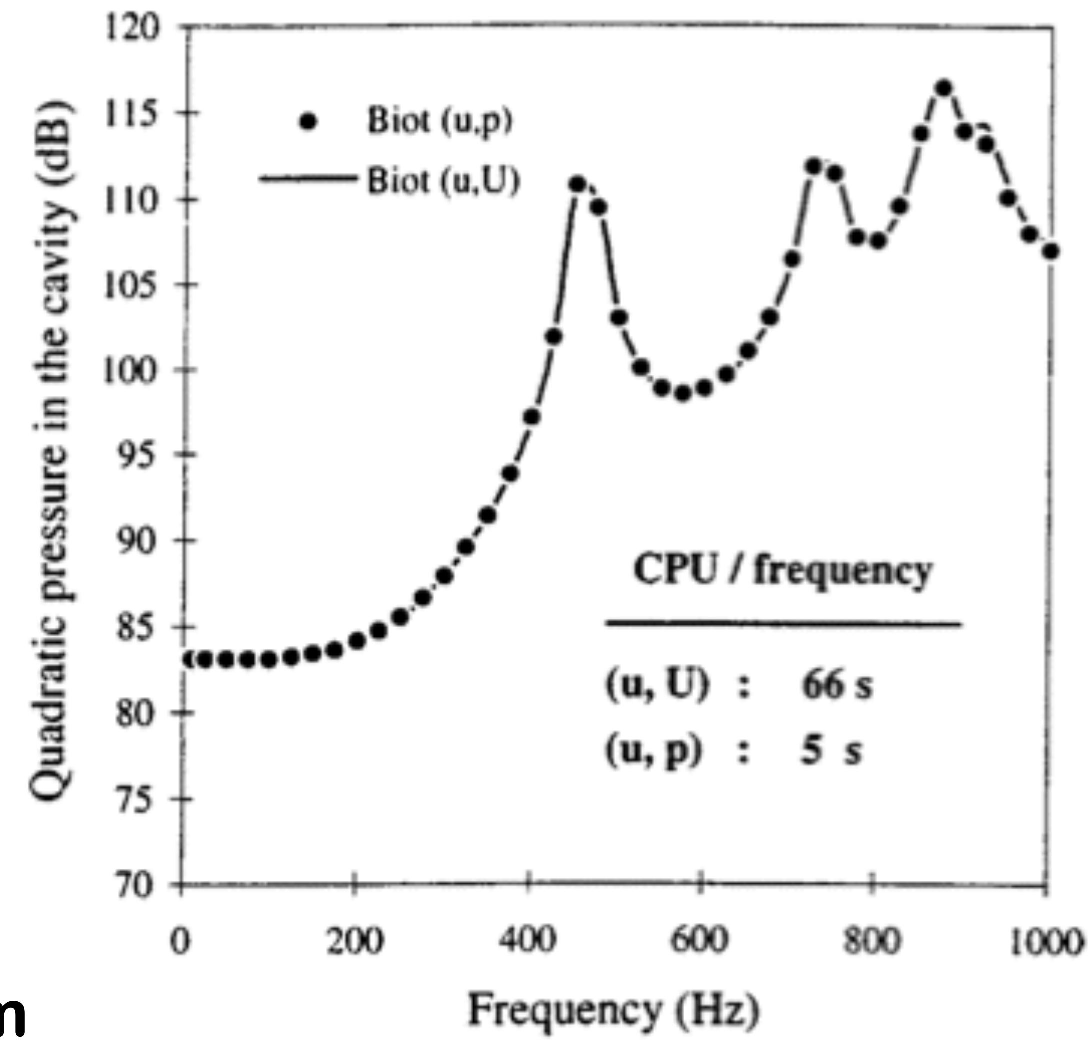
Atalla *et al.* JASA **104** 1998



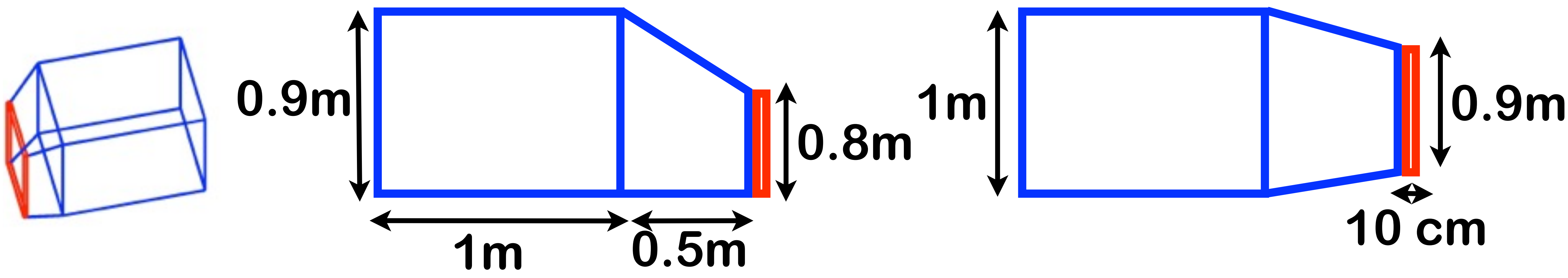
0.35x0.22x0.01 m cavity

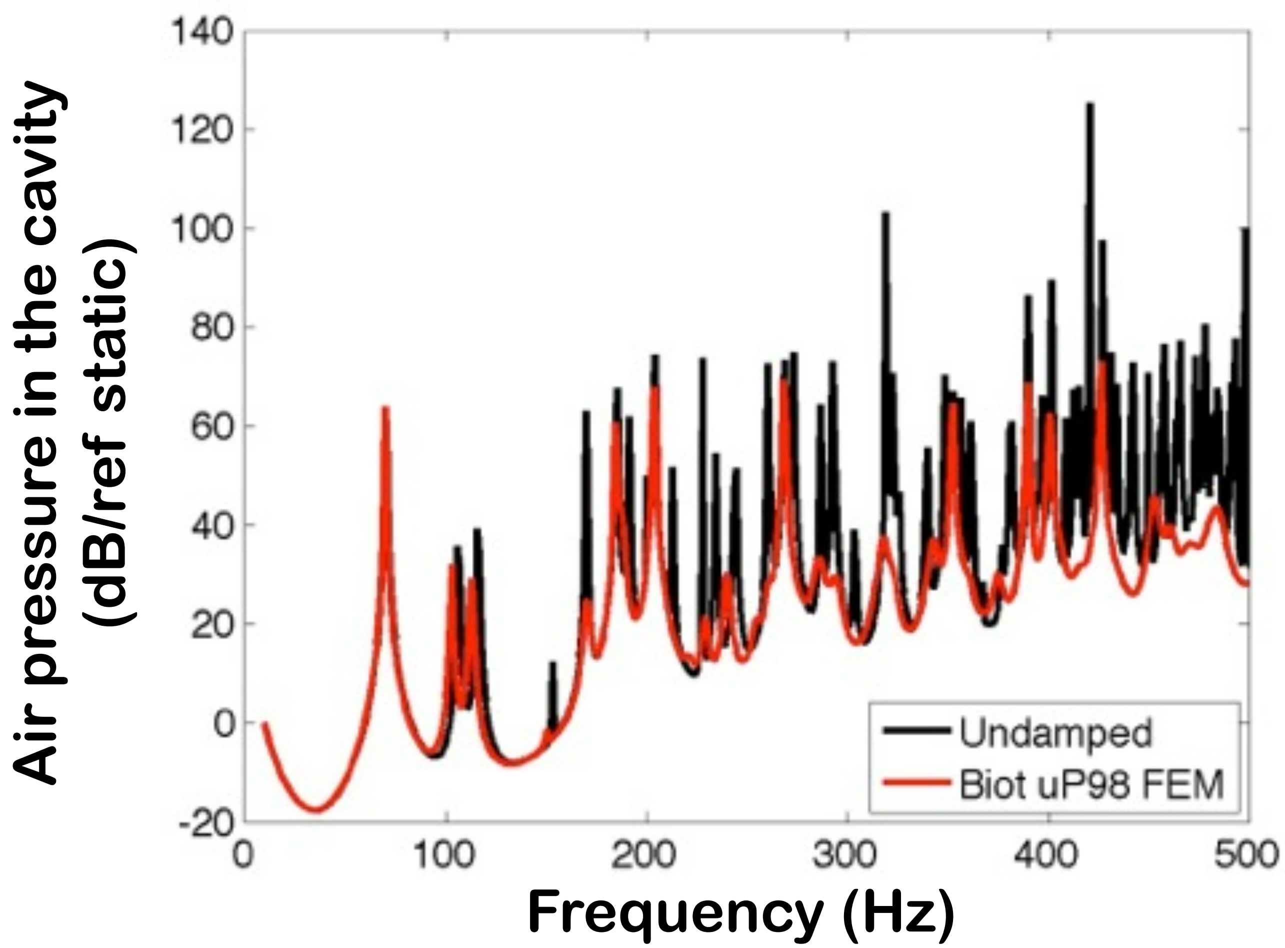


10 cm



A non cartesian geometry



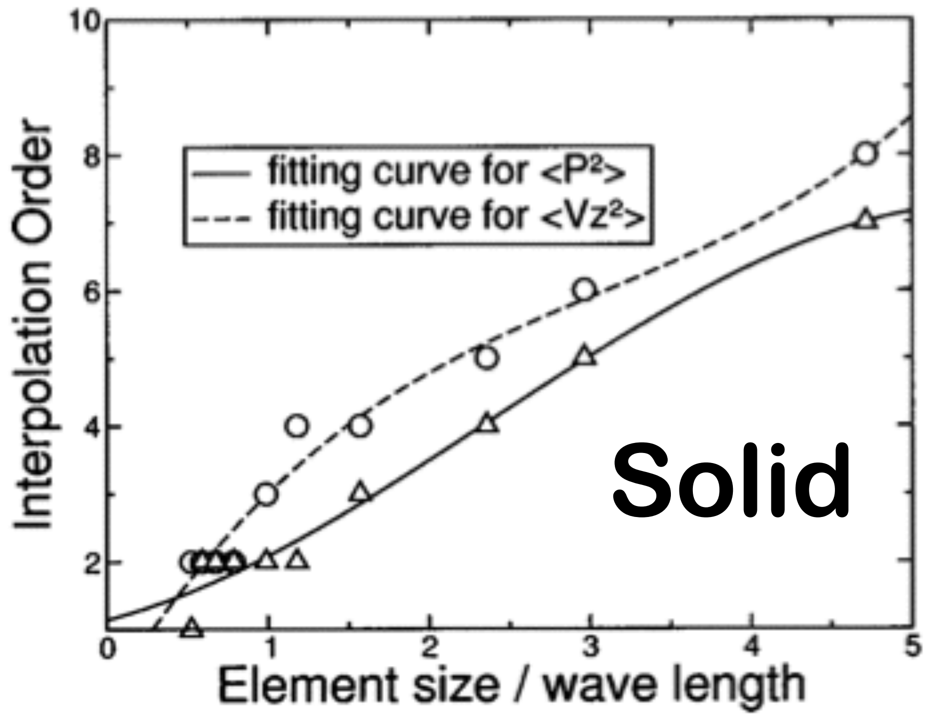


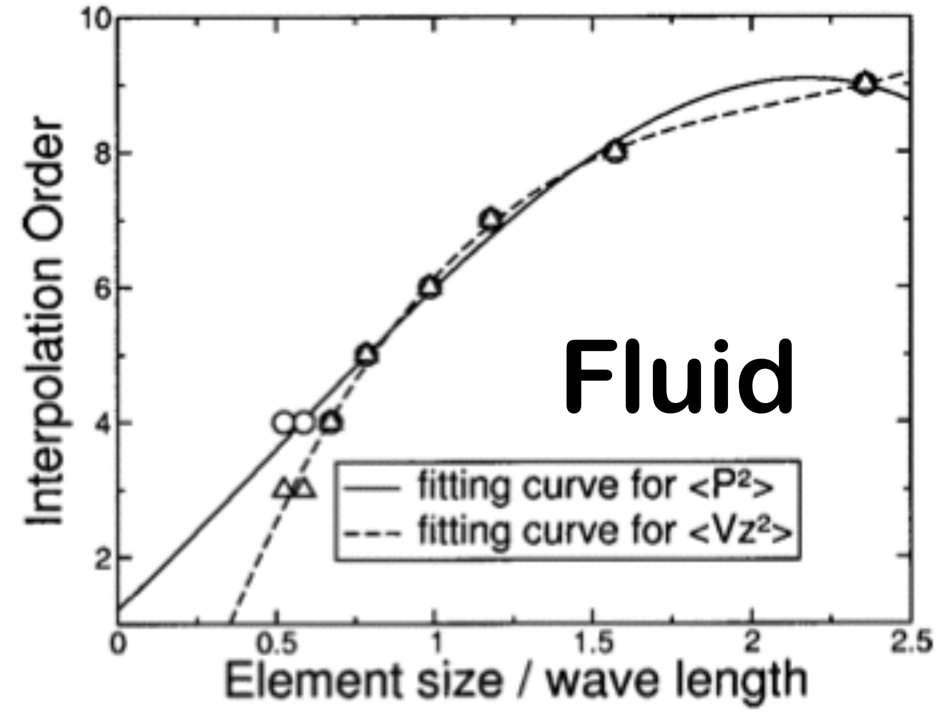
Home made Fortran code
Biot uP 1998
(35+5x8x7) HEXA27
500 Frequencies

Biot uP HSL ME57	18 802 s
Limp HSL ME57	3 012 s

Convergence

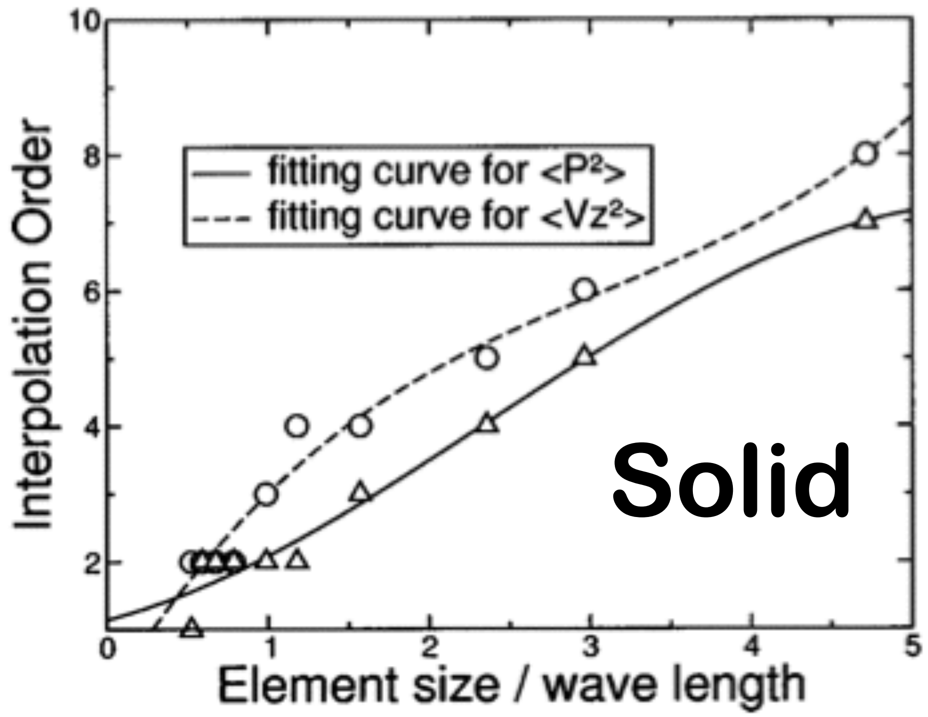
Rigobert *et al.* JASA **114(5)** 2003

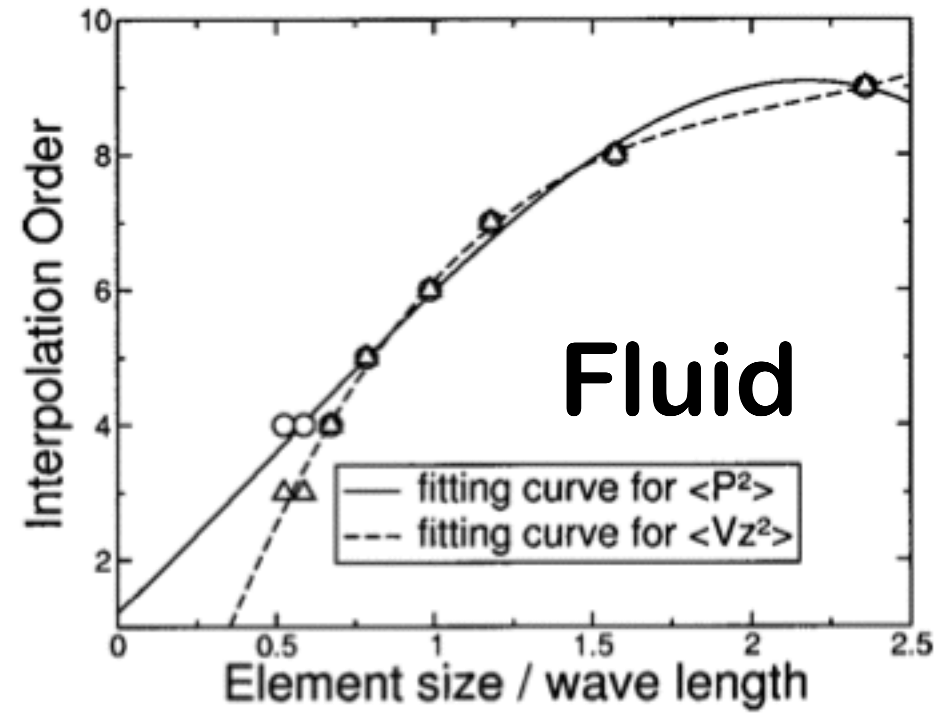


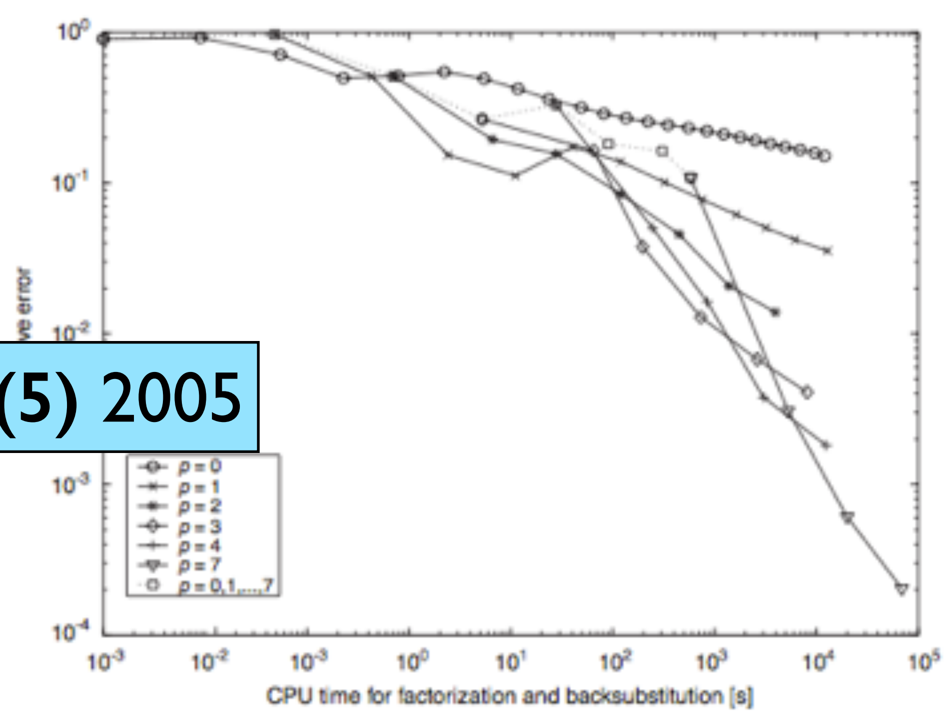


Convergence

Rigobert *et al.* JASA **114(5)** 2003



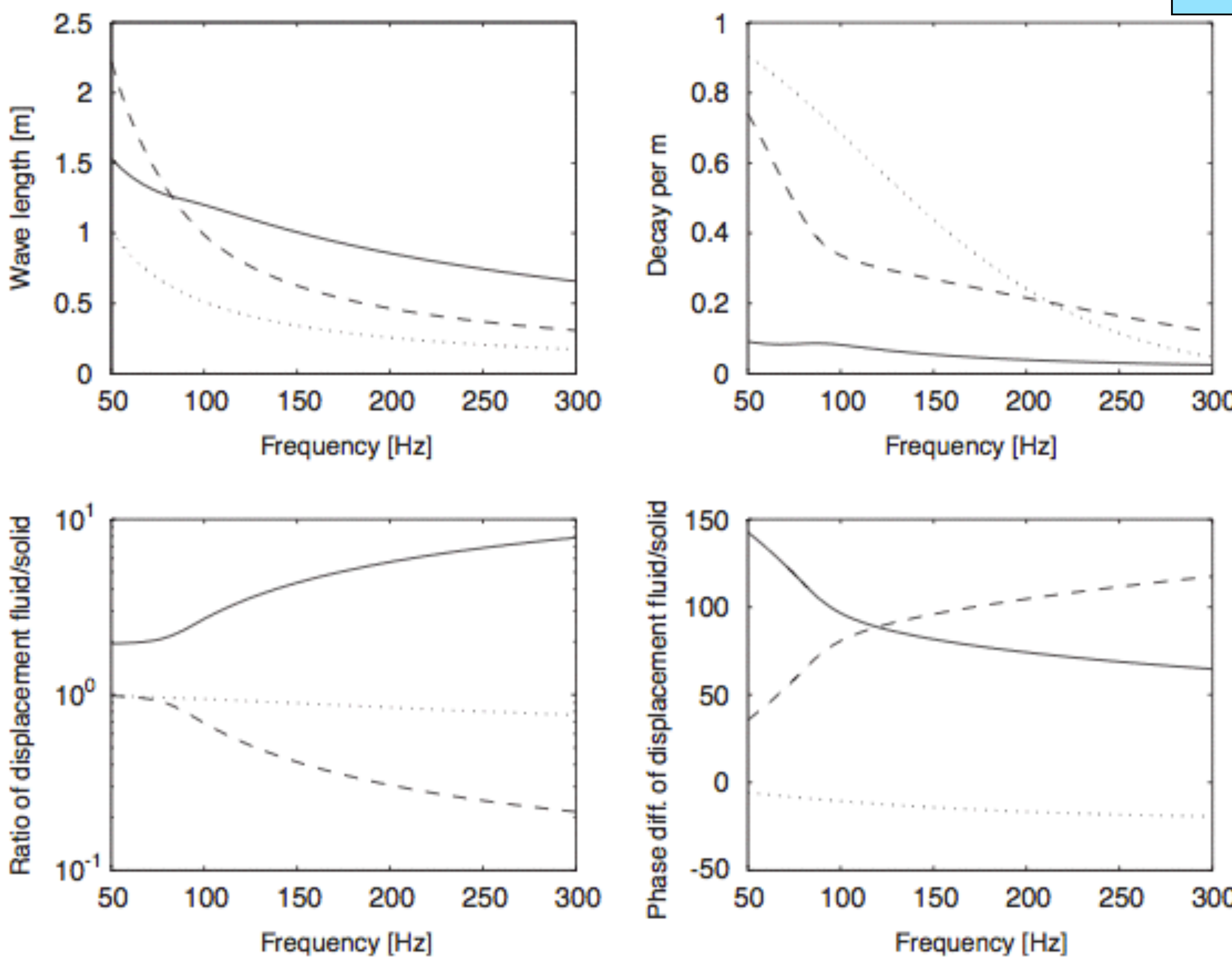




Hörlin JSV **285(5)** 2005

Problems relative to Biot waves (physics)

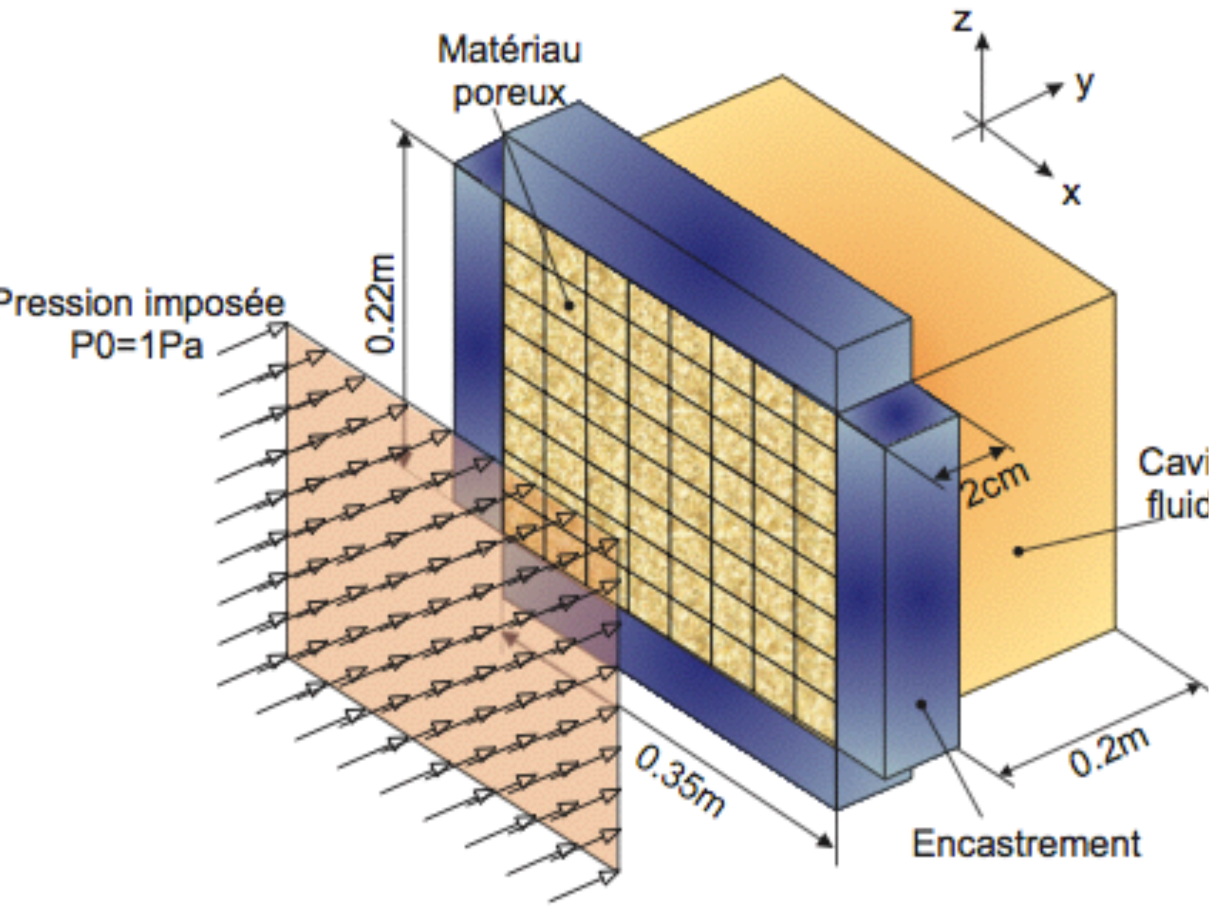
Hörlin JSV **285(5)** 2005

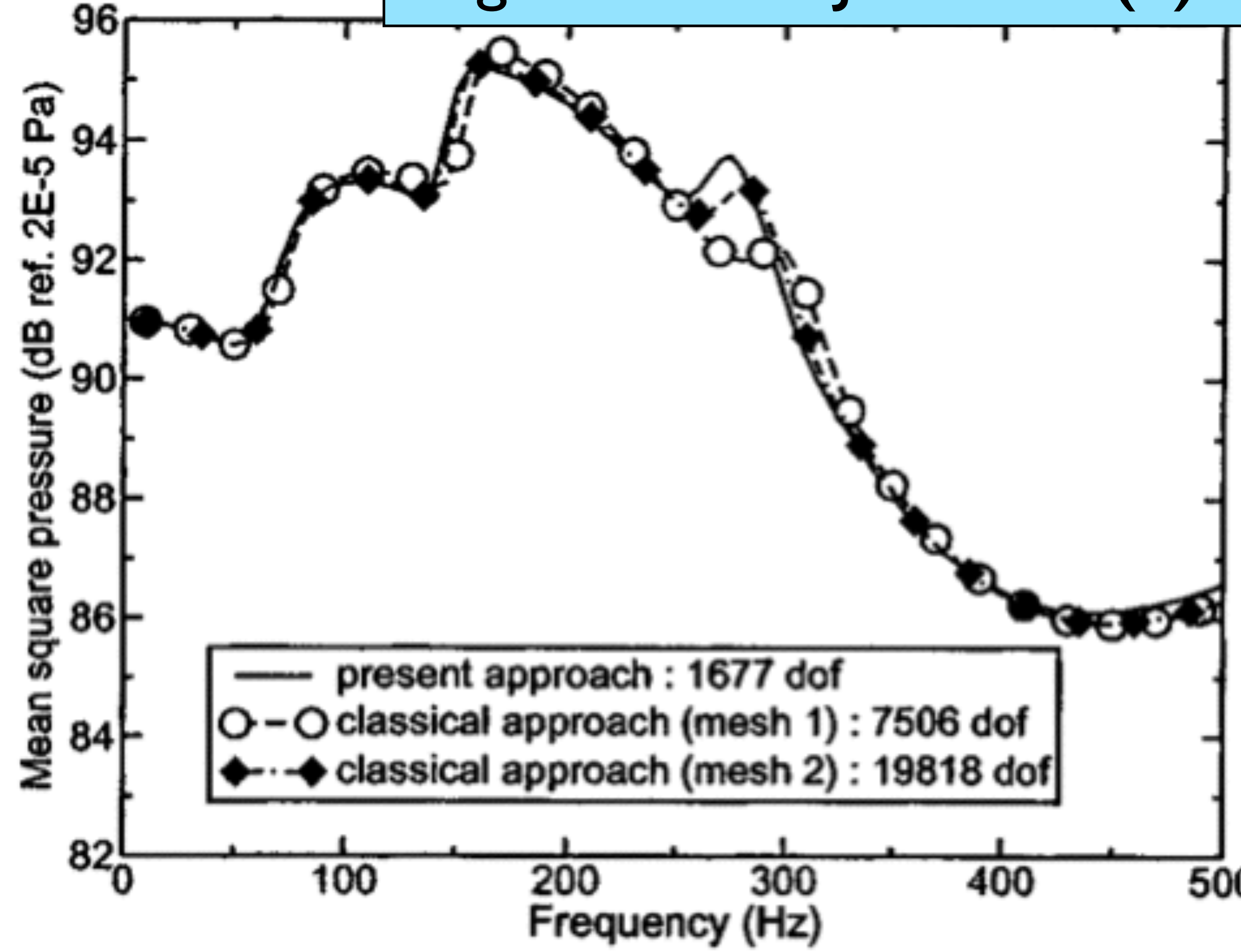


. 1. Wave properties for the air-borne compression wave (solid line), the frame-borne compression wave (dashed) and the shear wave (dotted line) versus the frequency. (The decay rate is close to one for a lowly damped wave.)

Hierarchical elements

Rigobert *et al.* JASA **115(6)** 2004

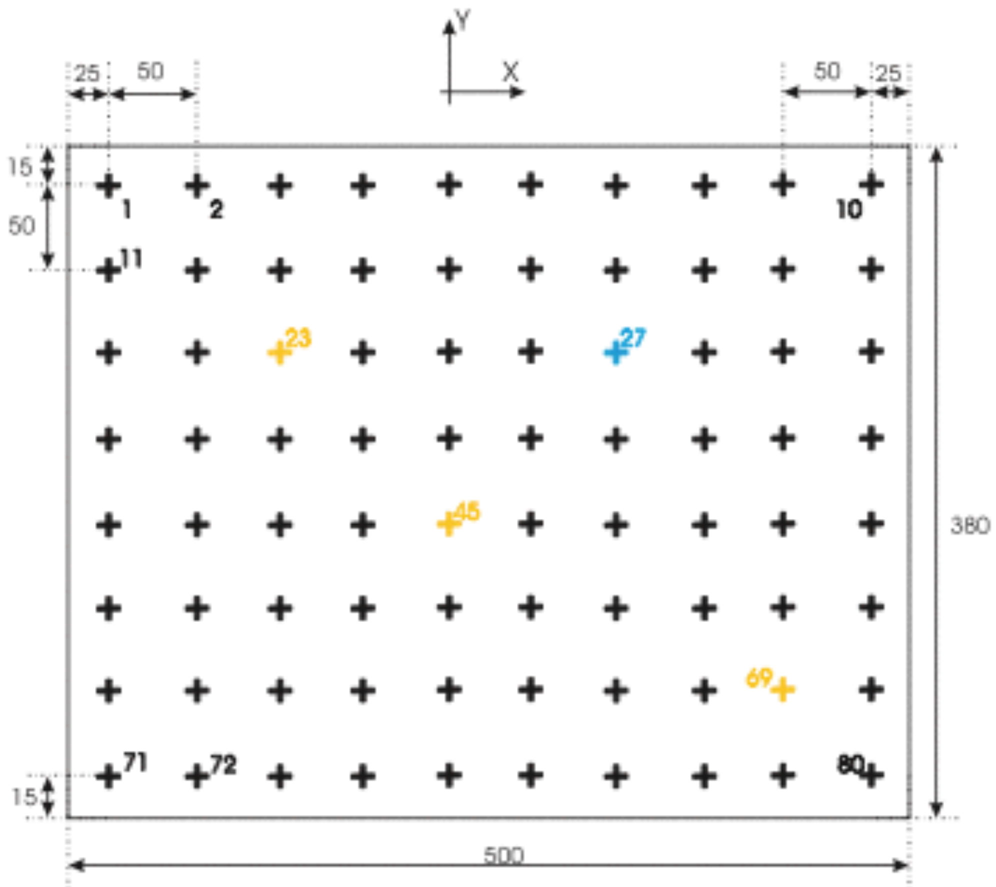




optimize the number of d.o.f. From a physical viewpoint, this indicates that a good prediction of the pressure levels in the cavity is closely related to a good description of the solid phase. The peak noticed in Fig. 6 at 275 Hz and associated

Validation case

Rigobert *et al.* JASA 115(6) 2004



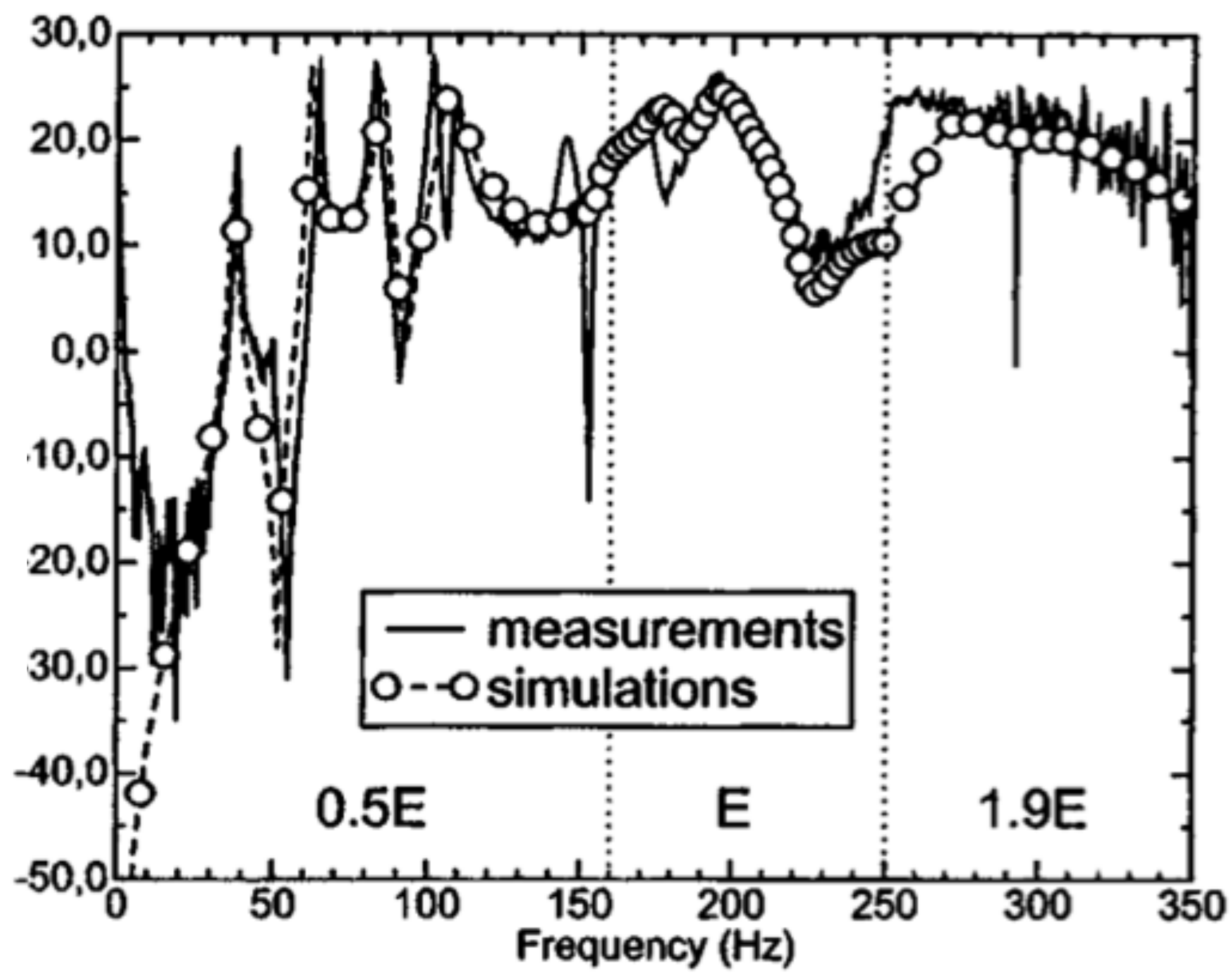
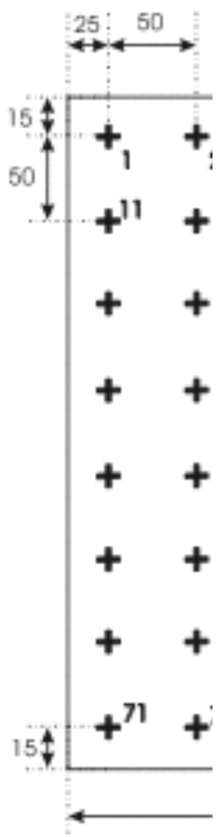
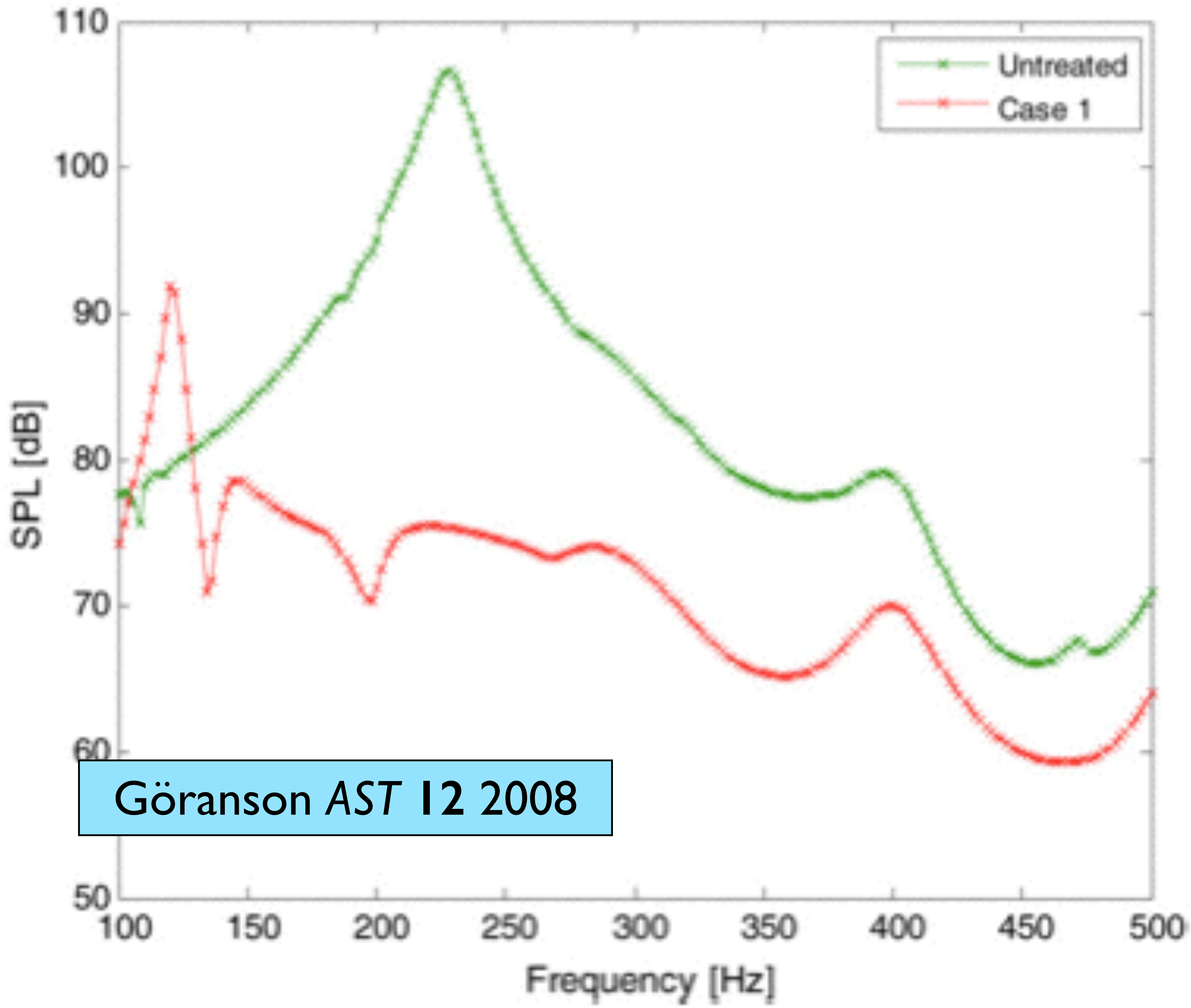


FIG. 8. Clamped plate coated with a polymer foam. FRF of the acceleration at point 69. Comparison between measurement and simulation.

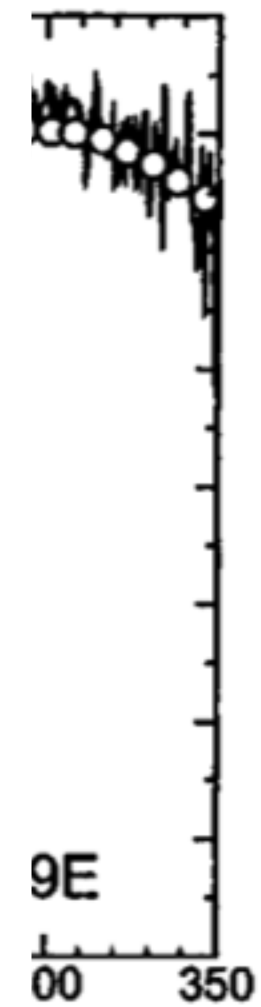
Validation case

Rigobert *et al.* JASA **115(6)** 2004





Göranson *AST* **12** 2008



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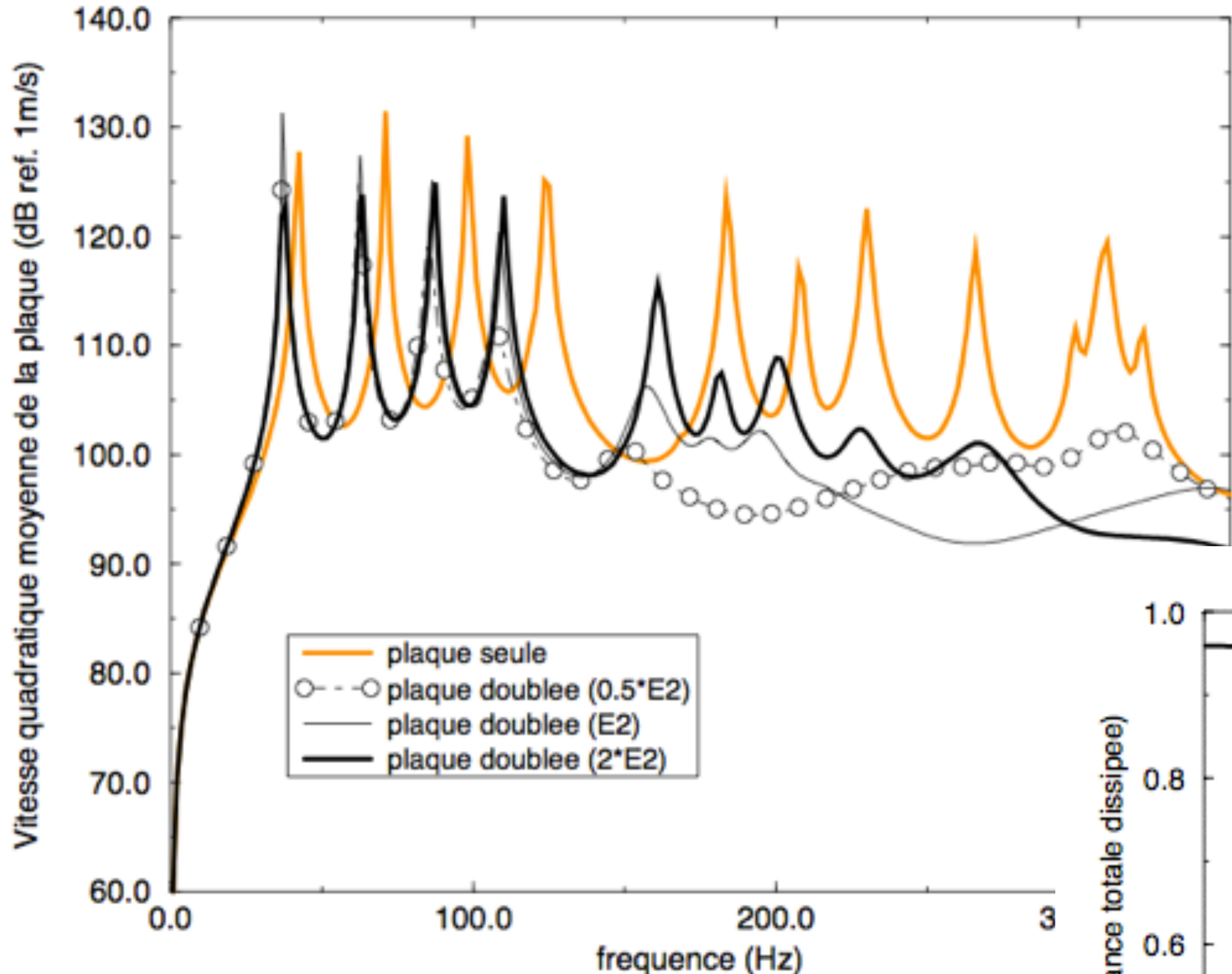
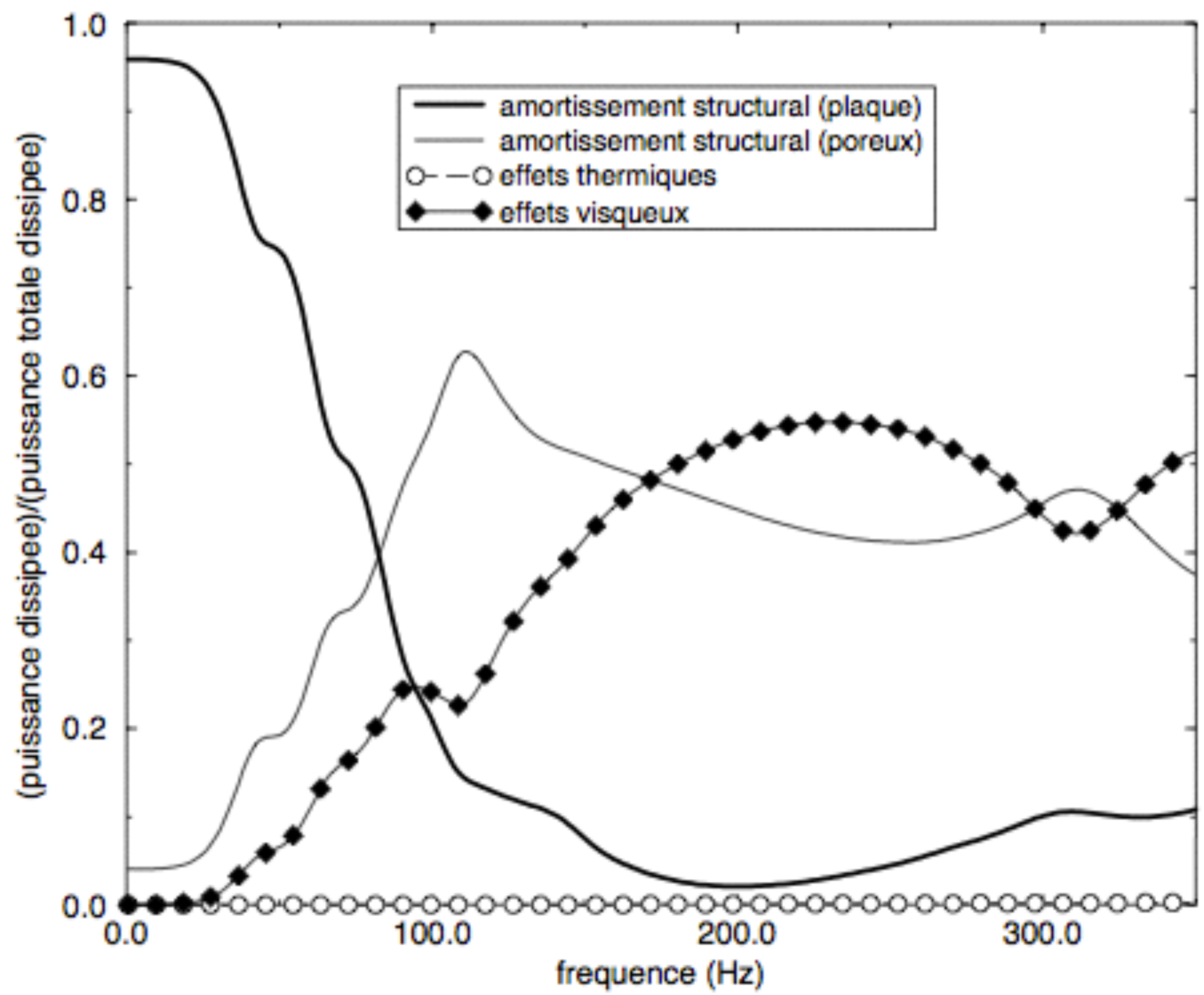
Influence of damping mechanisms

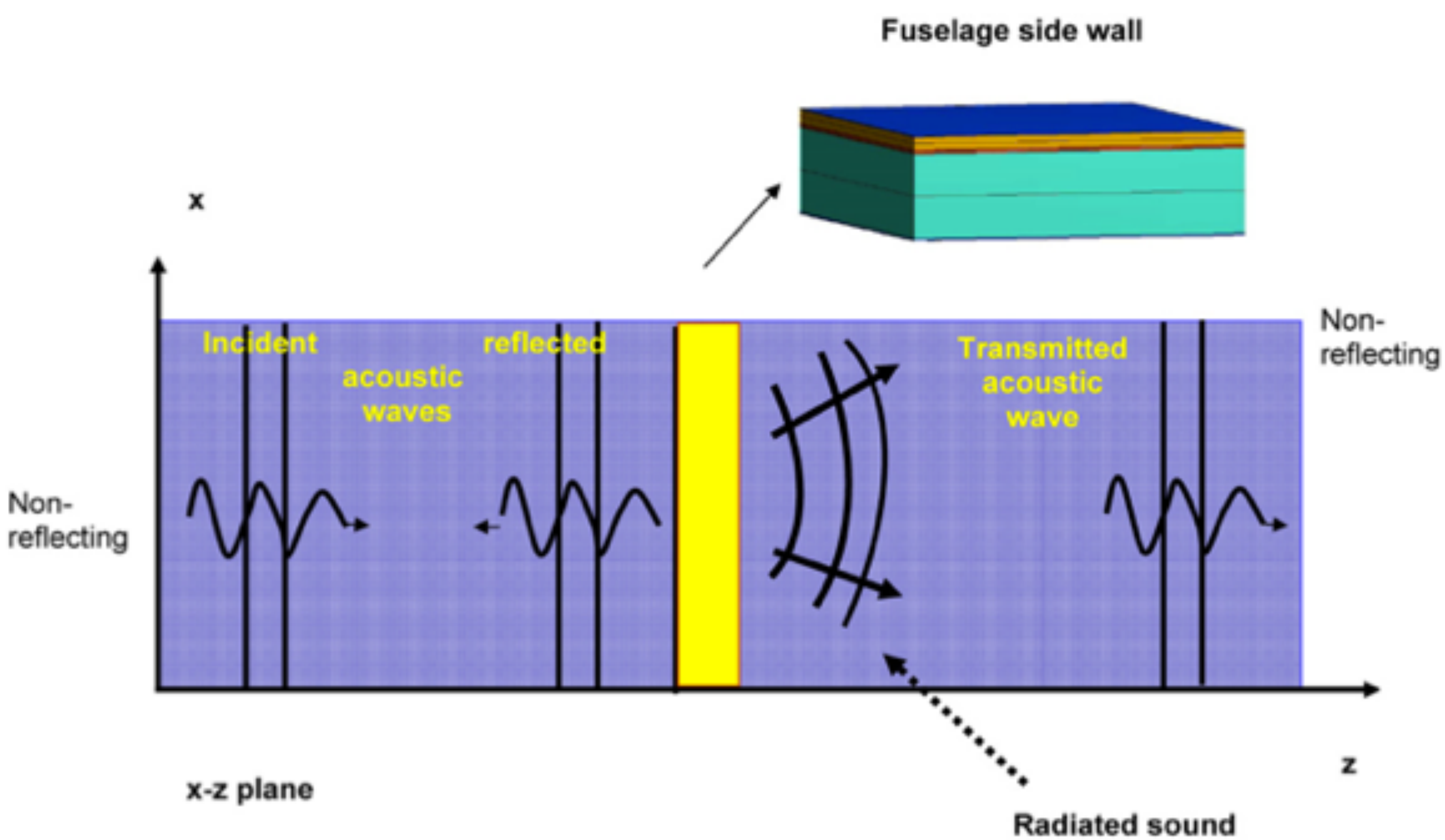




Influence of boundary conditions

It is obvious from the above discussions, that the boundary conditions between the panels and the foams introduced tend to be very important. In the cases where both panels are bonded to

Göranson *AST* **12** 2008



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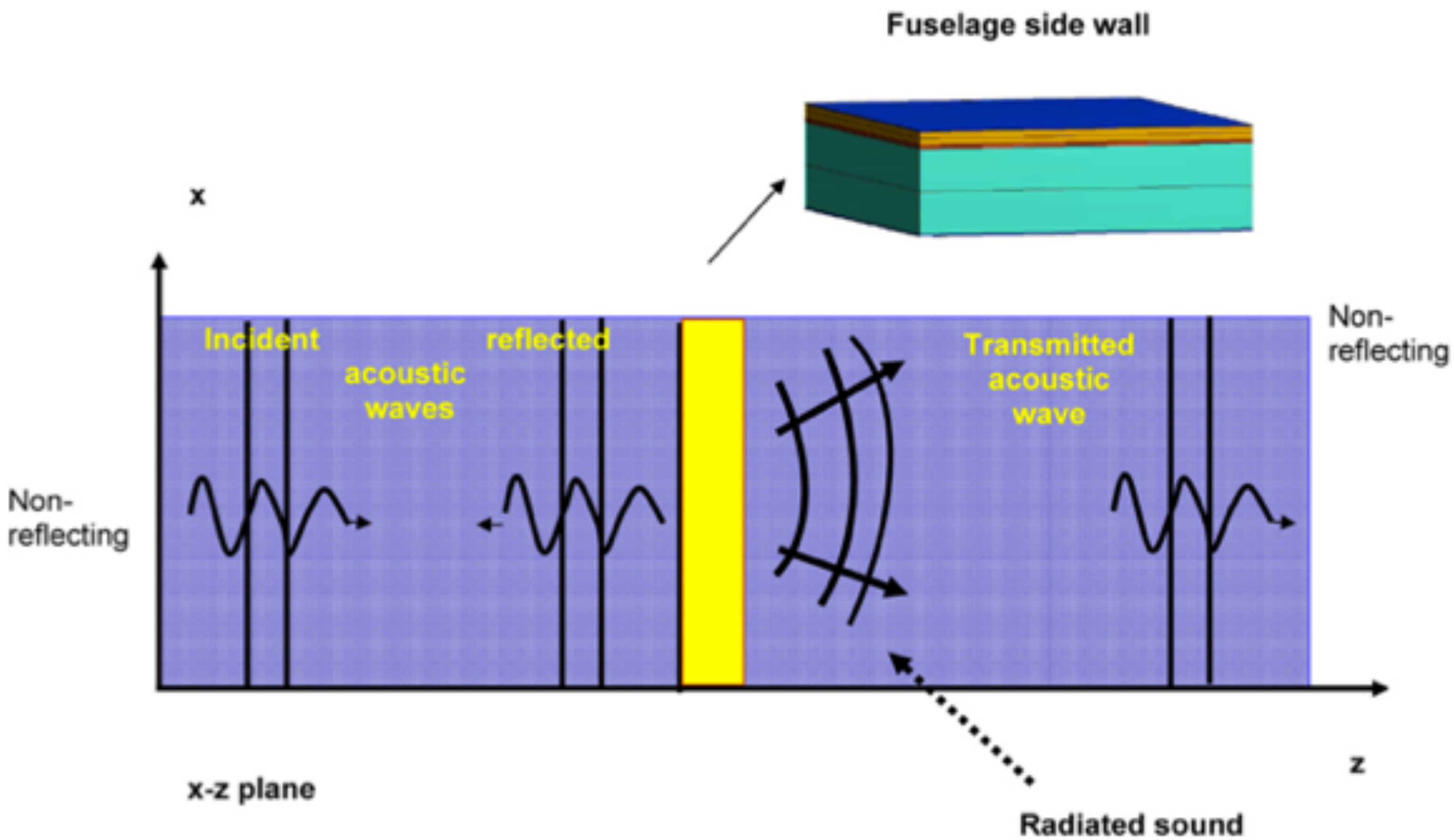

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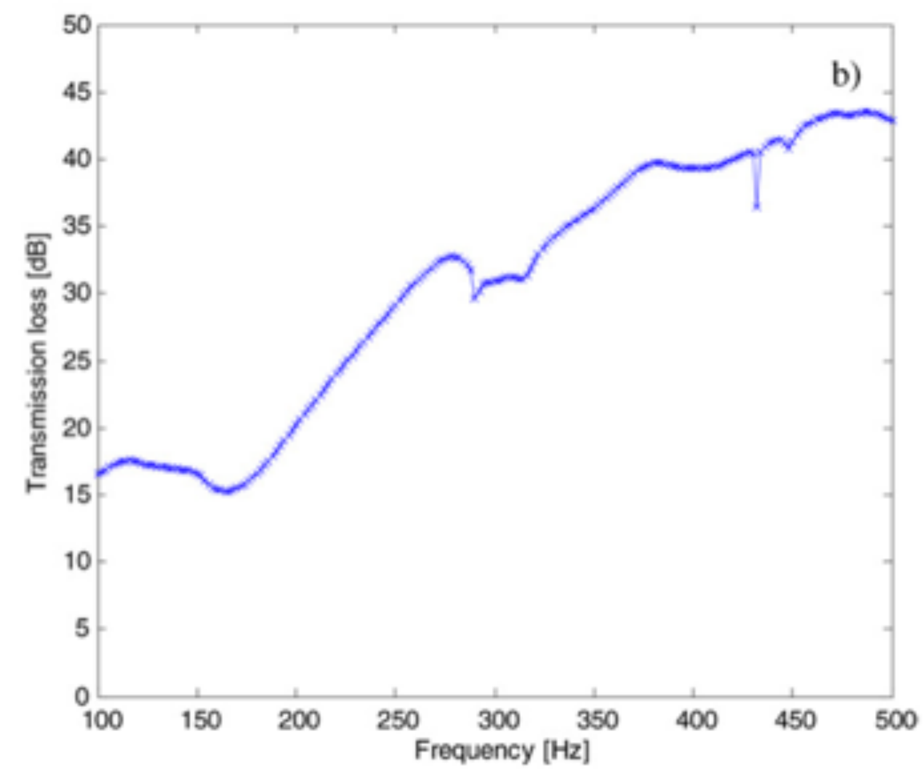

Influence of boundary conditions

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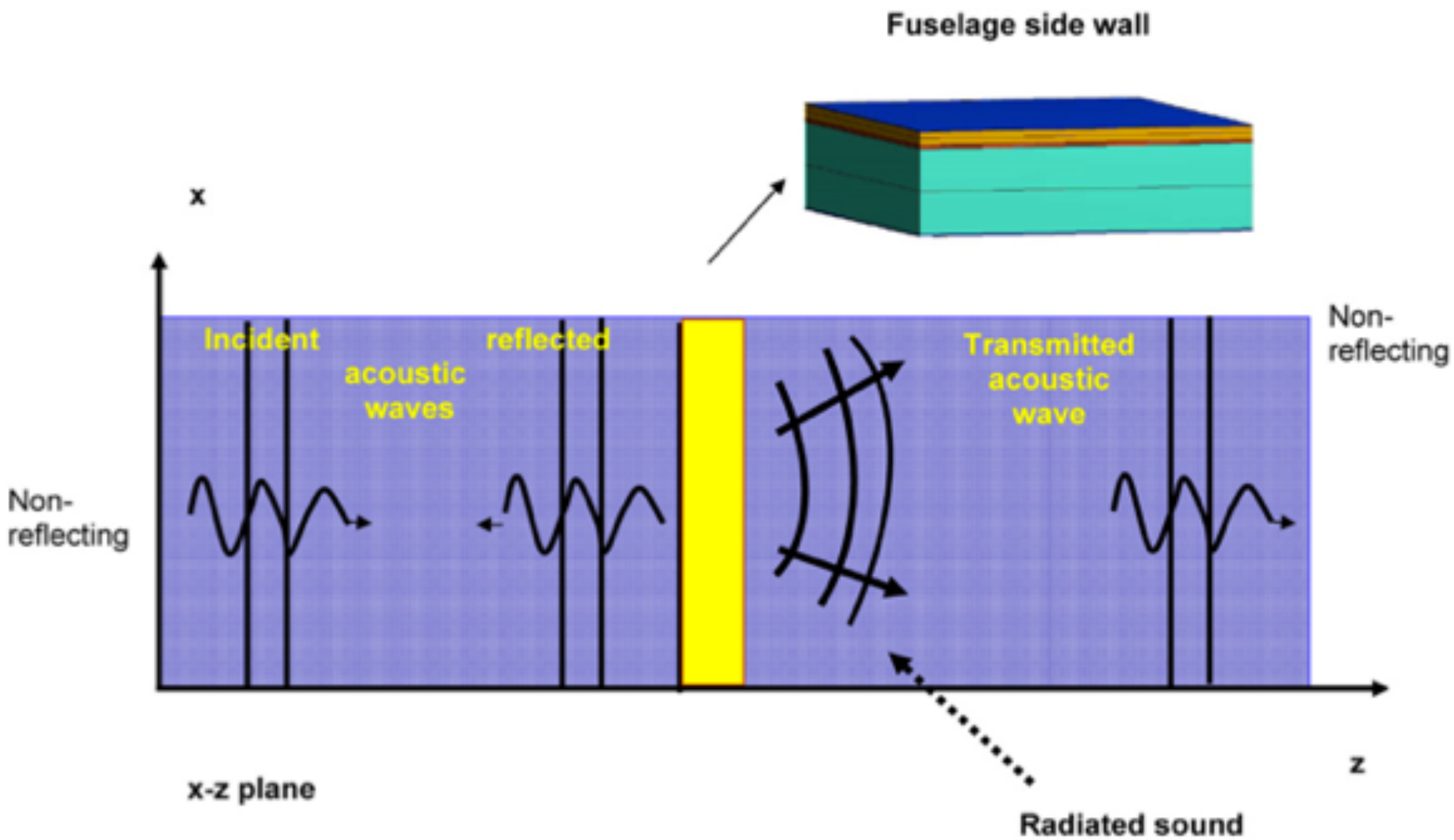
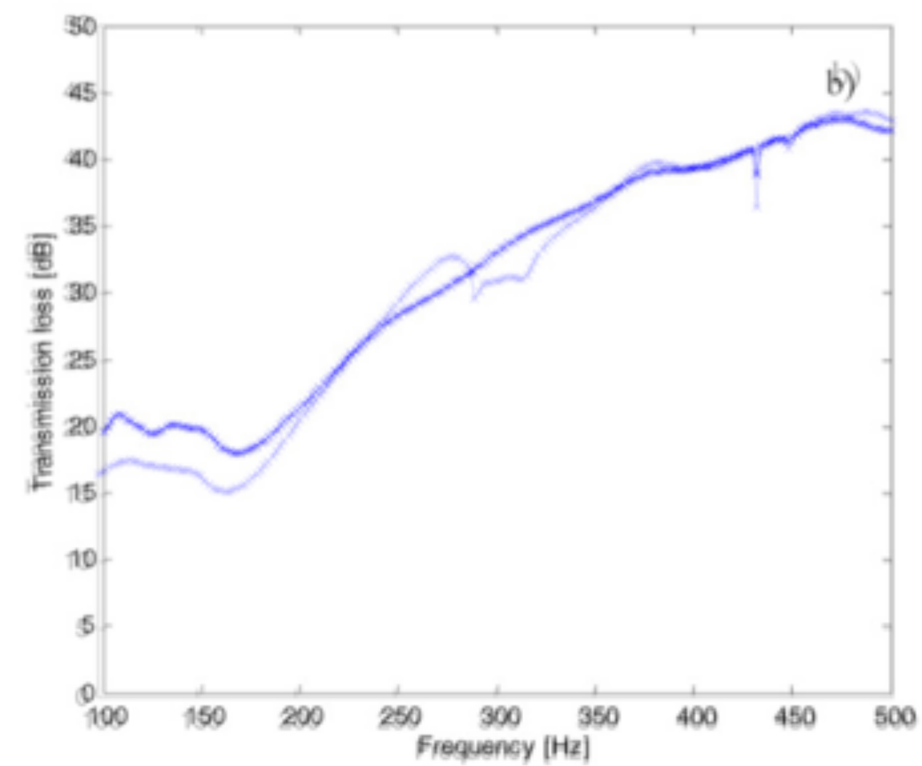
Göranson *AST* **12** 2008





Influence of boundary conditions

It
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Non-
reflecting